TokTickIT

CPE 334 Lab 4 — Actions Taken, Dashboards and Final Regression

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Reviewer: **Wisit Suwannao** (`Palapluem`)
Repository: github.com/N0TAW00D/TokTickIT
Branch of record: `main` (release #115, merge `d8f2de5`)

Answer Part 1

Git use with engineering workflow

Workflow. `lab4-staging` was cut from `main` after Lab 3 was released. Each of the nine Lab 4 Issues (#92–#100) was implemented on its own feature branch and merged into `lab4-staging` through a peer-reviewed pull request (#101–#114); `lab4-staging` reached `main` through one release pull request, #115, which closed Issues #93–#100 on merge. Nothing was committed directly to `main` or `lab4-staging`, and every pull request was reviewed and merged by `Palapluem`.

- Commit history: github.com/N0TAW00D/TokTickIT/commits/main
- Pull requests: all PRs
- Reviewer record: `docs/lab-04/reviewer.md`

Feature branch → `lab4-staging` → `main` evidence. First-parent history of `main` (`git log --first-parent --format='%h %cd %s' --date=short`; dates are committer dates in UTC+7), newest first; the Lab 4 release #115 sits on Lab 3's final #91:

```
d8f2de5 2026-11-03  #115 lab4-staging
99158e6 2026-10-04  #91 docs/lab-03-submission-polish
844f093 2026-10-04  #90 docs/lab-03-final
6d66399 2026-10-04  #89 lab3-staging
8d699e4 2026-09-12  #65 docs/20-submission-final-audit
ce2c9de 2026-09-08  #62 docs/20-submission-pdf
77e3142 2026-09-08  #64 lab2-staging
3396c41 2026-08-08  #12 lab1-staging
5d2ca71 2026-08-04  first commit
```

First-parent history of `lab4-staging` since Lab 3's last commit on `main` (`git log --first-parent 99158e6..lab4-staging`), newest first. Each line is a reviewed feature or fix branch merged into `lab4-staging`:

```
7027e74 2026-11-03  #114 docs/100-submission
4ccb5a2 2026-11-02  #113 docs/100-final-results
55dcf3c 2026-11-02  #112 feat/100-integration
f4ec16d 2026-11-02  #111 feat/99-hardening
78b6ba7 2026-10-31  #110 feat/96-actions-taken-ui
12f921e 2026-10-30  #109 feat/98-dashboard-ui
bfce69f 2026-10-30  #108 feat/95-ticket-workflow
4759377 2026-10-29  #107 fix/94-review-followups
732a058 2026-10-18  #106 feat/97-dashboard-api
d0a797c 2026-10-15  #105 feat/94-actions-taken-api
d748eb6 2026-10-14  #104 fix/93-seed-review
ef80e88 2026-10-13  #103 feat/93-actions-taken-model
89c2a7b 2026-10-13  #102 docs/92-spec-numeric-order
a7c9f92 2026-10-12  #101 docs/92-lab-04-spec
```

Commit graph. `git log --graph --format='%h %s' 99158e6..main` (subjects cut at 52 characters): every Lab 4 feature or fix branch leaves `lab4-staging`'s line, carries its commits, and returns through its merge commit; the top commit is the release merge #115 into `main`.

```
* d8f2de5 Merge #115 lab4-staging
*   7027e74 Merge #114 docs/100-submission
|\
| * a4185e2 docs(lab-04): write the review log and t
|/
*   4ccb5a2 Merge #113 docs/100-final-results
|\
| * 306801a docs(lab-04): record the final test resu
|/
*   55dcf3c Merge #112 feat/100-integration
```

```
|\
| * d8aed9d test(workflow): create the fixture users
| * 82f593e test(e2e): expect the stale status draft
| * 9b0d439 test(e2e): capture Lab 4 screenshots fro
| * bbe0dbb fix(client): keep status count badge and
| * 1fd81a6 test(e2e): capture Lab 4 screenshots at
| * 778eac4 test(e2e): cover Actions Taken, resoluti
| * 5bba67b fix(client): keep Actions table row butt
|/
*   f4ec16d Merge #111 feat/99-hardening
|\
| * 6d879aa test(e2e): follow the ui-spec field orde
| * c8f356a fix(test): give the other-routes authori
| * 7101601 docs(lab-04): state the order of recentl
| * b9089ad docs(lab-04): drop UNIT-08 now that the
| * df5cd66 refactor(server): remove unused resoluti
| * b4ba372 test(lab-04): cover reopened and latest-
| * 09ba42e fix(client): allow a future time when ca
| * b8c937a fix(actions): allow cancelling a future-
| * 0325011 docs(lab-04): let a scheduled Planned ac
| * f723fab test(e2e): add E2E-17 console/links/plac
| * b1ddbee test(e2e): add E2E-16 safe failure in th
| * bdf1016 test(e2e): remove the hardening fixtures
| * f551683 test(e2e): add Lab 4 accessibility rows
| * d8b1e45 fix(client): make the link in the resolu
| * 903abf2 test(e2e): add Lab 4 responsive rows R-0
| * d67a9df fix(client): keep dashboard status count
| * ce7716e docs: say the Requester sees every Actio
| * 1db4c20 docs: add Lab 4 test map and per-role de
| * d6b5a36 test(api): cover history, workflow, dash
| * 1a7974b test(api): add dashboard and Actions lis
| * d7af321 test(api): add route inventory guarding
|/
*   78b6ba7 Merge #110 feat/96-actions-taken-ui
|\
| * f34578f test(client): name the review-round styl
| * 2e7e71a docs(lab-04): add the #110 review-round
| * b9e3dce fix(client): use error tokens for form e
| * aa18d3e fix(client): follow ui-spec field pairs,
| * 7b14eec fix(client): put Save action above Cance
| * a0c11b7 refactor(client): trim branch comment bl
| * dea6901 refactor(client): share one invalidateRe
| * 0220177 test(client): stub the Lab 4 reads in th
| * ba9d0c6 fix(client): announce Status history loa
| * 5550534 fix(client): ignore late Actions Taken a
| * d564c30 fix(client): ignore late Actions Taken a
| * 12cfe1e docs(lab-04): record the Requester detai
| * 279572d fix(client): wait for the appended entry
| * f8032e7 fix(client): wait for effect-driven focu
| * 01e6ba7 fix(client): wait for thread entries aft
| * c6f40ee test(client): cover Action badge, token
| * cff0469 test(client): count the Actions Taken an
| * 9058c24 feat(client): read-only Actions Taken an
| * b8befa1 fix(client): await deferred focus restor
| * b37ecfb test(client): cover actions taken edit,
| * ef4010e test(client): cover actions taken create
| * ed5619a feat(client): actions taken card with li
|/
*   12f921e Merge #109 feat/98-dashboard-ui
|\
| * 9ce063c docs(lab-04): add R-06 for the drill-dow
| * 997dcdb test(e2e): measure drill-down chip remov
| * 5d7698c test(client): cover the drill-down chip
| * d7d3b6f fix(client): show a visible Remove label
| * 922d6a0 docs(lab-04): add the C-33 role filter h
| * 080e815 test(client): C-33 the chosen role survi
```

```
| * 801d4d3 fix(client): keep the User Management ro
| * a6a5e60 docs(lab-04): record the final list of D
| * 7bb80fe test(e2e): open My Tickets after the rem
| * 5876734 test(e2e): open My Tickets after switchi
| * ca22ab6 docs(lab-04): add the requester-switch s
| * 77772d7 fix(client): let My Tickets filters wrap
| * 5395f6c test(e2e): open My Tickets explicitly af
| * 87cb7ad docs(lab-04): list the post-login naviga
| * f789aa4 fix(client): let the header nav wrap so
| * 20ffdd9 test(e2e): move post-login landing and f
| * 63745f9 feat(client): hide owner, status and com
| * c844538 feat(client): read dashboard drill-down
| * c22ff2d test(client): cover dashboard token disc
| * 6be6e37 feat(client): add Requester dashboard an
| * 77c76aa feat(client): add IT Staff dashboard scr
| * 79eb0cd feat(client): make Dashboard the first n
|/
*   bfce69f Merge #108 feat/95-ticket-workflow
|\
| * b7ac442 fix(client): keep unsaved workflow input
| *   144f07c Merge lab4-staging > feat/95-ticket-wo
| |\
| * | 993492a test(workflow): cover every role, curr
| * | d8c89d3 test(client): assert the attempted val
| * | 5895723 fix(client): keep the attempted status
| * | ae0b111 test(lab-03): send version and seed a
| * | 5c97308 test(client): cover ticket workflow ro
| * | 5b3233a feat(client): version-guarded ticket w
| * | 29f5a11 feat(workflow): guard owner and IT Pri
| * | e5fe487 test(lab-03): send version on owner an
| * | aed1eb9 test(lab-02): expect version in the ti
| * | f9c7fcd docs(lab-04): keep the Lab 3 404 for a
| * | 5864439 test(seed): send version on the status
| * | 9f47297 feat(workflow): guard status changes w
| * | 6ab804f test(lab-03): send version on status P
| * | e06b62d feat(workflow): add pure ticket transi
* | |   4759377 Merge #107 fix/94-review-followups
|\ \ \
| |_|/
|/| |
| * | d22dccf docs(lab-04): spell out that attachmen
| * | e2f6b09 docs(lab-04): state that attachmentNot
| * | e4a7038 test(actions): pin the attachmentNotes
| * | 531ef8f refactor(actions): drop the offset ran
| * | ec17ca6 docs(lab-04): record the empty-JSON-bo
| |/
* |   732a058 Merge #106 feat/97-dashboard-api
|\ \
| |/
|/|
| *   ebd0cd5 Merge lab4-staging > feat/97-dashboard
| |\
| |/
|/|
* |   d0a797c Merge #105 feat/94-actions-taken-api
|\ \
| * | 7e733eb test(actions): cover malformed-JSON or
| * | a70fbb7 fix(actions): reject impossible calend
| * | d7fc3c9 fix(actions): parse the JSON body afte
| * | 7afe760 test(actions): pin BR-34 precedence, s
| * | 6538512 test(actions): cover UUID v4 and the s
| * | 03c5af4 fix(actions): validate body shape befo
| * | 623b902 test(actions): cover Actions Taken aut
| * | 1d78f14 feat(actions): add Actions Taken list,
| * | 11de4d3 feat(actions): add pure Action Taken v
|/ /
```

```
| * a0c36c2 feat(dashboard): add GET /api/dashboard/
| * 8192b4c feat(dashboard): add GET /api/dashboard/
| * 6f2c144 feat(tickets): apply dashboard drill-dow
| * a350a21 docs(lab-04): list the widened My Ticket
| * 9eea7f4 feat(validation): parse dashboard drill-
| * e8b18f8 feat(dashboard): add Bangkok window and
|/
*   d748eb6 Merge #104 fix/93-seed-review
|\
| * 4bf6668 docs(lab-04): trace MIG-06b-06e and BR-1
| * 176fdf9 docs: say re-seeding skips Tickets that
| * 6d5db3b fix(db): treat requester-resolved and du
| * ee2731f fix(db): repair status history left by t
| * 8eb7f64 fix(db): keep attachments, comments and
| * a48e18d fix(db): hash a valid original create bo
| * 3d16267 fix(db): keep application activity on re
| * c6bb81d feat(actions): add canonical create-body
|/
*   ef80e88 Merge #103 feat/93-actions-taken-model
|\
| * 2a5225e test(lab-04): cover the Lab 4 seed with
| * 4ac2c07 feat(db): extend seed with Actions Taken
| * cf59f31 docs: document Lab 4 migration recovery
| * ed20785 test(lab-04): cover Lab 4 migration with
| * 06ed4ce test(db): add db:test:lab3-fixture datab
| * c9559d6 feat(db): add Lab 4 down-migration recov
| * 9db5b4e feat(db): add Actions Taken, status hist
* |   89c2a7b Merge #102 docs/92-spec-numeric-order
|\ \
| |/
|/|
| * f7fab70 docs(lab-04): list business rules and ac
|/
* a7c9f92 Merge #101 docs/92-lab-04-spec
* 065c01c docs(lab-04): order contract tables numeri
* 9c7ecea docs(lab-04): add an Evidence column to th
* aa99ac2 docs(lab-04): reject a reused clientReques
* cf0d122 docs(lab-04): state the lock, read, compar
* 0f4b3ef docs(lab-04): require Result only to compl
* b5720be docs(lab-04): list every earlier test the
* 64a8e43 docs(lab-04): test 415-before-401 and the
* 34da0da docs(lab-04): put the 415 content-type che
* aeda4bf docs(lab-04): add dump/restore, forbidden-
* 117e77e docs(lab-04): dashboard forbidden state, u
* f31ca21 docs(lab-04): align API check order, actio
* 5098ad3 docs(lab-04): fix check order, actionAt se
* fe7ea7b docs(lab-04): list every Lab 2/3 assertion
* 024d02f docs(lab-04): gate closing on open Actions
* cb4badd docs(lab-04): serialize status changes and
* 7f43477 docs(lab-04): cross-reference fixes after
* b89805d docs(lab-04): list every changed Lab 3 con
* 8c9a894 docs(lab-04): write the Sprint 4 test plan
* 1fe9b1f docs(lab-04): write the Sprint 4 UI specif
* 8594f2a docs(lab-04): write the Sprint 4 API contr
* 678320c docs(lab-04): write the Sprint 4 specifica
* 71af372 docs(lab-04): scaffold the Sprint 4 contra
```

Project board (Kanban). GitHub Project: TokTickIT Individual Sprints (columns Backlog → Specified → Started → PR Review → (Fixing) → Done). Final state, read from the board itself with `gh project item-list 3 --owner N0TAW00D --limit 100 --format json` on **2026-11-02 17:36 UTC** (after the release #115): the board holds **30 items, every one in Done** (Backlog 0, Specified 0, Started 0, PR Review 0, Fixing 0, Done 30), and every one of the 30 Issues is closed. The Lab 4 items are #92–#100 (#92 closed when #101 merged; #93–#100 closed by #115). Complete table view of the board (Board = the item's column; Issue = the GitHub Issue state):

#	Issue	Board	Issue state	#	Issue	Board	Issue state
#1	Set up the TokTickIT project foundation	Done	Closed	#69	Role-based authorization & role-aware application shell	Done	Closed
#2	Implement the API health check	Done	Closed	#70	Requester regression on authenticated identity, Public Comments & Problem Appears Resolved	Done	Closed
#3	Create and seed IT request categories	Done	Closed	#71	IT Staff Ticket Queue — queue API & responsive UI	Done	Closed
#4	Display the IT request category list	Done	Closed	#72	IT Staff Ticket Detail — ownership, IT Priority, status workflow & Internal Notes	Done	Closed
#13	Lab 2 sprint specification and test plan	Done	Closed	#73	Administrator User Management — minimalist user administration	Done	Closed
#14	Database schema, seed data & automated-test database	Done	Closed	#74	E2E, responsive & visual inspection, and release integration	Done	Closed
#15	Development Requester context, app shell & Zen Green foundation	Done	Closed	#92	Lab 4 sprint specification and test plan	Done	Closed
#16	Create Ticket — reference data, creation API & form UI	Done	Closed	#93	Actions Taken data model, migration & Lab 4 seed data	Done	Closed
#17	Attachment lifecycle — upload, download & soft removal	Done	Closed	#94	Actions Taken API — create, update, authorization & stale-update handling	Done	Closed
#18	My Tickets — Requester-scoped list API & responsive UI	Done	Closed	#95	Ticket workflow — final transition matrix, resolution gate & conflict handling	Done	Closed
#19	Requester Ticket Detail — owned-ticket API & read-only view	Done	Closed	#96	Actions Taken UI on Ticket Detail — list, create & view/edit modes	Done	Closed
#20	Responsive & visual inspection, E2E flow & release integration	Done	Closed	#97	Role dashboards API — Requester & IT Staff metrics with drill-down	Done	Closed
#66	Lab 3 sprint specification and test plan	Done	Closed	#98	Role dashboards UI — metric cards, recent Tickets & dashboard navigation	Done	Closed
#67	User model, Lab 2 migration & Lab 3 seed data	Done	Closed	#99	Final regression, accessibility & product hardening	Done	Closed
#68	Authentication foundation — hashing, login/logout/current-user & mandatory password change	Done	Closed	#100	E2E, responsive & visual inspection, and release integration	Done	Closed

Reviewer record (`docs/lab-04/reviewer.md` §1–§4). Author `N0TAW00D` (Natthawat Primsirikunawut); reviewer `Palapluem` (Wisit Suwannao), who reviewed and merged every Lab 4 pull request; `THN4` was not a Lab 4 reviewer. “Rounds” counts review cycles (each changes-requested or finding-raising review plus the final approval).

PR	Title	Review outcome	Rounds
#101	Lab 4 specification, API spec, UI spec and test plan (#92)	Changes requested (gate vs regression rule) → approved	2
#102	BRs and ACs in strict numeric order (#92)	Approved	1
#103	Actions Taken model, migration, seed (#93)	Changes requested ×4 findings; merged before they were fixed	1
#104	Seed review fixes (#93)	Changes requested ×3 → approved	4
#105	Actions Taken API (#94)	Changes requested ×2 → approved	3
#106	Dashboards API (#97)	Approved; approved again after conflict merge	2
#107	#105 follow-ups (#94)	Changes requested → approved	2
#108	Ticket workflow and resolution gate (#95)	Changes requested ×2 → approved	3
#109	Dashboards UI and Dashboard landing (#98)	Inline findings, changes requested → approved	2
#110	Actions Taken UI (#96)	Changes requested (6 inline) → approved	2

#111	Final regression, accessibility, hardening (#99)	Approved	1
#112	E2E, screenshots, visual fixes (#100)	Commented, changes requested (server timeouts) → approved	3
#113	Final test results in `tests.md` (#100)	Approved	1
#114	Review log and AI use record (#100)	Approved	1
#115	`lab4-staging` → `main` release	Approved	1

Representative review findings and resolutions (full text and replies in `reviewer.md` §2–§3):

#101	The strict resolution gate (BR-17) would break two earlier tests while the plan claimed it listed every edit — the gate was kept (D-27) and both re-pointings listed in `tests.md` §4.2.
#103–#104	Reseeding after real activity (a status change, an upload, “problem appears resolved”) rewrote Ticket state — the seed now leaves such Tickets alone; MIG-06b–06e prove each case.
#105	Validation order (BR-34): access and `STALE_UPDATE` were checked before field validity — body shape first, then access, then merged-state validation, then the version compare.
#108	Unsaved workflow input was lost after failures and Refresh (BR-36) — drafts for status, owner and IT Priority now survive and are never resubmitted automatically.
#109	Role filter not in the URL; the chip’s “×” button was 28 px on mobile — URL-backed filter (C-33); labelled 44 × 44 px target (R-06).
#110	A late response for ticket A could replace ticket B’s rows — both detail screens now drop reads for a ticket that is no longer current (C-36).
#112	The full server suite timed out under load (10 s hook in WF-11) — the fixture users are created once per file: that file 627 s → 102 s, the suite 1363 s → 512 s, all 1175 passing; no timeout was raised.

One process exception is recorded rather than hidden: #103 was merged by the reviewer while its changes-requested review was still open; the four findings were fixed in #104 and re-reviewed in four rounds.

README and `.gitignore`. `README.md` (prerequisites, one-command bootstrap, databases, seeded accounts, server / client / e2e / full-suite commands, the Lab 4 test map and per-role demo walk-through, project structure) and `.gitignore` (excludes `memory/`, `server/uploads/`, `.env*` except the `.example` templates, `node_modules`, build output).

Repository structure (top level; verified in the working tree):

```
client/              React + TypeScript + Vite frontend       client/tests/lab-04/
server/              Express + TypeScript + Prisma backend    server/tests/lab-04/
e2e/                 Playwright E2E + responsive + evidence   e2e/lab-04/
docs/lab-04/         specification, api-spec, ui-spec, tests, reviewer, ai-use, submission
artifacts/lab-04/    screenshots/{staff-dashboard, requester-dashboard,
                       actions-taken, ticket-workflow}
README.md  .gitignore  docker-compose.yml  package.json
```

Answer Part 2

Specification-driven development

Link: `docs/lab-04/specification.md` companions `docs/lab-04/api-spec.md`, `docs/lab-04/ui-spec.md`, `docs/lab-04/tests.md`. The specification holds Functional Requirements FR-01–FR-23 (§4), an authorization matrix (§4.1), Business Rules BR-01–BR-41 (§5), the Ticket status transition matrix (§5.1), data changes with migration, backfill and recovery (§7), 27 numbered design decisions D-01–D-27 (§7.6, §11), Acceptance Criteria AC-01–AC-52 (§9) with FR → AC coverage, and the Definition of Done (§10).

Sample numbered contract items (full lists at the link):

FR-01	IT Staff and Administrators can create an Action Taken under a Ticket with action date/time, description, result, optional assignee, follow-up flag and note, and attachment notes; the backend records the performer.
FR-04	An Action Taken moves through a status lifecycle (Planned → In Progress → Completed; Cancelled from either non-terminal state); Completed and Cancelled are read-only.
FR-10	The backend refuses Resolved unless the resolution gate (BR-17) holds, returning the reasons.
FR-12	Ticket status, owner and IT Priority writes, and Action Taken updates, are rejected as stale when the client's `version` is no longer current, and the response carries the current resource.
BR-01	An Action Taken belongs to exactly one Ticket, fixed at creation.
BR-02	The Ticket Owner coordinates the Ticket, but an Action Taken may be performed by a different IT Staff member.
BR-17	**Resolution gate.** A transition to Resolved requires (a) a Ticket Owner, (b) at least one Completed Action Taken, and (c) no Action Taken in Planned or In Progress; otherwise `RESOLUTION_BLOCKED`, listing every failed condition. Enforced by the backend for any client.
BR-22	**Recent window:** time zone `Asia/Bangkok` (UTC+7, no DST); `windowStart` is 00:00 Bangkok on the day six days before today; "updated recently" means `Ticket.updatedAt >= windowStart`.
AC-01	Given a permitted IT Staff user and valid data, when an Action Taken is created, then it is saved under the correct Ticket with the authenticated creator as performer and the approved assignee.
AC-20	Given a Ticket with no Completed Action, when Resolved is requested directly through the API, then it is rejected with `RESOLUTION_BLOCKED` naming that reason.
AC-27	Given seeded Tickets for a Requester, when the Requester dashboard is retrieved, then Open, Waiting for Me, Recently Updated and Recently Resolved equal the values recomputed directly from the database.

Authorization matrix (`specification.md` §4.1; "Own" = records the user owns as Requester; ‡ = new or changed in Lab 4):

Operation	Requester	IT Staff	Administrator
Log in / out, read own identity, change own password	✓	✓	✓
Create Ticket	✓	—	—
List / view Tickets	Own	All	All (read)
Post Public Comment	Own	All	—
IT Staff Ticket Queue (read) ‡	—	✓	✓
Claim / assign / reassign owner	—	✓	—
Change Ticket status (incl. Resolved, Closed, Cancelled)	—	✓	—
Read Actions Taken ‡	Own	All	All
Create / update Action Taken ‡	—	✓	✓
Read Ticket status history ‡	Own	All	All
Requester dashboard ‡	✓	—	—
IT Staff dashboard ‡	—	✓	✓
User counts inside the dashboard ‡	—	—	✓

List / create / edit users, set initial password	—	—	✓

Ticket status transition rules (`specification.md` §5.1; every transition is IT Staff only, unchanged from Lab 3, with the BR-17 gate on every arrival at Resolved; a pair not listed, including a no-op, is `INVALID_TRANSITION`):

From	Permitted next	Notes
New	Open, In Progress, Cancelled	In Progress needs an owner
Open	In Progress, Waiting for Requester, Resolved, Cancelled	Resolved needs BR-17
In Progress	Waiting for Requester, Resolved, Cancelled	Resolved needs BR-17
Waiting for Requester	In Progress, Resolved, Cancelled	In Progress needs an owner; Resolved needs BR-17
Resolved	Closed, Reopened	Closed needs BR-40 (no open Actions)
Closed	Reopened	No Action writes (BR-12)
Reopened	In Progress, Waiting for Requester, Resolved, Cancelled	Resolved needs BR-17
Cancelled	—	Terminal; cancelling also cancels open Actions (BR-41)

Dashboard calculations (`specification.md` BR-20–BR-24, `api-spec.md` §4–§5): all metrics are computed by the backend from the database in the `Asia/Bangkok` time zone. The Requester dashboard returns open Tickets, Tickets waiting for the Requester, recently updated and recently resolved Tickets for the caller's own Tickets only. The IT Staff dashboard returns unassigned Tickets, Tickets owned by the caller, counts by status and by IT Priority (open Tickets), urgent and recently updated Tickets, follow-ups, and the caller's open Actions Taken; an Administrator also gets user counts. Each card has a drill-down query that returns the same count (API-22, API-28). Lists are capped at five items with a "View all" link; empty states are documented in `ui-spec.md` §3–§4.

Migration decisions (`specification.md` §7.4, §7.6): the Lab 4 migration adds an `ActionTaken` table, an append-only `TicketStatusChange` table and `Ticket.version` without discarding any Lab 2 or Lab 3 row, with a documented recovery path that is tested on a Lab 3-shaped database (MIG-01..MIG-09). Justified decisions include **D-07** (an integer `version` column instead of comparing `updatedAt`, because BR-15 bumps `updatedAt` on Action writes and would report false conflicts), **D-03** (a separate append-only history table instead of deriving history, so "append-only" is structural) and **D-20** (follow-up consistency as a database `CHECK` as well as service validation). Legacy Tickets get an empty history rather than invented rows (D-10). The seed is idempotent and leaves Tickets with real application activity alone (MIG-06b–06e).

Product Definition of Done (`specification.md` §10.1) — the coding agent may claim "done" only when: every FR is implemented and every BR enforced by the backend; every AC passes with an automated test listed in `tests.md`; the authorization matrix is enforced server-side, never by hiding a control; the migration applies to a Lab 3 database with everything intact and the seed is idempotent; the full suite (unit, API/integration, UI component, UI style, responsive, accessibility, security/authorization, migration/regression, E2E) passes with nothing skipped, disabled or `.only`; every major screen is verified at desktop, tablet and mobile with the visual checklist complete; and nothing from the excluded list (§3.2) is built. §10.2 adds the staging → main workflow, Kanban all Done, `reviewer.md`, `ai-use.md`, this before-implementation evidence, and a current README and `.gitignore`.

Evidence the specification existed before the implementation. From the repository history:

```
$ git log --reverse --format='%ci  %h  %s' -- docs/lab-04/specification.md | head -2
2026-10-12 14:53:10 +0700  71af372  docs(lab-04): scaffold the Sprint 4 contract documents and
repository structure
2026-10-12 15:03:50 +0700  678320c  docs(lab-04): write the Sprint 4 specification
$ git log --reverse --format='%ci  %h  %s' -- docs/lab-04/api-spec.md docs/lab-04/ui-spec.md docs/
lab-04/tests.md
2026-10-12 15:05:08 +0700  8594f2a  docs(lab-04): write the Sprint 4 API contract
2026-10-12 15:07:06 +0700  1fe9b1f  docs(lab-04): write the Sprint 4 UI specification
2026-10-12 15:09:21 +0700  8c9a894  docs(lab-04): write the Sprint 4 test plan and AC traceability
```

```
$ git log -S'ActionTaken' --reverse --format='%ci  %h  %s' -- server/prisma/schema.prisma | head -1
2026-10-12 23:55:52 +0700  9db5b4e  feat(db): add Actions Taken, status history and Ticket version
(Lab 4 migration)
```

The four contract documents were written on **2026-10-12 between 15:03 and 15:09 +0700** and merged to `lab4-staging` through #101 at **16:20 +0700**. The first Lab 4 implementation commit (the migration) is at **23:55 +0700 the same day**, and the first implementation pull request, #103, merged on **2026-10-13 at 10:16 +0700** — after the specification pull request. Every later implementation pull request (#105–#112) merged between 2026-10-15 and 2026-11-02.

Answer Part 3

Test-driven development and traceability

Link: `docs/lab-04/tests.md`. It holds the commands and rules (§1), the planned tests with a Final column (§2), the Acceptance Criterion → test matrix (§3), the Labs 1–3 regression rules and the permitted edits to earlier tests (§4) and the final output (§5).

Planned tests and actual test files (`tests.md` §1.1, §2; every row below has Final = Pass):

Level	IDs	Test files	Rows
Unit	UNIT-01..11	`server/tests/lab-04/{action-rules, action-request-hash, ticket-transitions, dashboard-window, …}.test.ts`	11
API / integration	API-01..35	`server/tests/lab-04/{actions-taken, requester-dashboard, staff-dashboard, …}.api.test.ts` `route-inventory.test.ts`	35
Workflow	WF-01..16	`server/tests/lab-04/ticket-workflow.api.test.ts`	16
Authorization / security	SEC-01..08	`server/tests/lab-04/authorization.api.test.ts`	8
Migration	MIG-01..09 (+06b–06e)	`server/tests/lab-04/migration.test.ts`	13
Performance smoke	PERF-01..02	`server/tests/lab-04/performance-smoke.api.test.ts`	2
UI component	C-01..37	`client/tests/lab-04/{StaffDashboard, RequesterDashboard, ActionsTaken, TicketWorkflow, …}.test.tsx`	37
UI style	S-01..09	`client/tests/lab-04/ui-style.test.tsx`	9
Responsive	R-01..06	`e2e/lab-04/{responsive, drilldown-chips}.spec.ts`	6
Accessibility	A11Y-01..06	`e2e/lab-04/accessibility.spec.ts`	6
End-to-end	E2E-01..19	`e2e/lab-04/{actions-taken-flow, ticket-resolution, dashboards, regression-smoke, screenshots}.spec.ts`	19
Regression (Labs 1–3)	REG-01..07	`server/tests/lab-0{1,2,3}/`, `client/tests/lab-0{1,2,3}/`, `e2e/lab-0{2,3}/`	7
Total		`npm run test:all`	**169**

AC traceability (`tests.md` §3): every one of AC-01…AC-52 maps to at least one test, for example AC-01 → API-01, E2E-01; AC-20 → WF-03, E2E-07; AC-24 → API-16, WF-08, WF-09, C-26, C-37, E2E-09; AC-27 → API-17, E2E-12. Authorization is asserted at the HTTP layer (SEC and WF rows), never by UI visibility.

Final status as recorded in `tests.md` §5 (one run of each suite on `lab4-staging` at `55dcf3c`, the code that #115 carried to `main` unchanged; #113 and #114 only changed documents; nothing skipped, disabled or `.only`):

```
npm run test:server   Test Files  40 passed (40)   Tests  1175 passed (1175)   485 s
npm run test:client   Test Files  31 passed (31)   Tests   552 passed (552)
npm run test:e2e      352 passed (7.8m)            one worker, database reset first
planned rows: 169 / 169 Pass (Labs 1–3 regression included as REG-01..07)
```

Complete passing output, file by file (from the per-test JSON of those runs; each number is the count of tests that passed in that file, and no test failed):

Server — 40 files, 1175 tests

`lab-01/app.test.ts`	2	`lab-01/categories.test.ts`	1
`lab-02/attachments.api.test.ts`	54	`lab-02/create-ticket.api.test.ts`	12
`lab-02/my-tickets.api.test.ts`	14	`lab-02/reference-data.api.test.ts`	4
`lab-02/seed.test.ts`	7	`lab-02/test-db-identity.test.ts`	11
`lab-02/test-db-runner.test.ts`	2	`lab-02/ticket-detail.api.test.ts`	8

lab-02/ticket-number.test.ts	9	lab-02/validation.test.ts	54
lab-03/attachment-download-staff.api.test.ts	7	lab-03/auth.api.test.ts	75
lab-03/authorization.api.test.ts	23	lab-03/comment-validation.test.ts	9
lab-03/comments-notes.api.test.ts	27	lab-03/migration.test.ts	8
lab-03/password.test.ts	14	lab-03/staff-assignable-users.api.test.ts	11
lab-03/staff-queue.api.test.ts	24	lab-03/ticket-detail-staff.api.test.ts	11
lab-03/ticket-it-priority.api.test.ts	17	lab-03/ticket-notes.api.test.ts	20
lab-03/ticket-owner.api.test.ts	21	lab-03/ticket-status.api.test.ts	88
lab-03/users-admin.api.test.ts	58	lab-04/action-request-hash.test.ts	4
lab-04/action-rules.test.ts	36	lab-04/actions-taken.api.test.ts	49
lab-04/authorization.api.test.ts	16	lab-04/dashboard-window.test.ts	7
lab-04/migration.test.ts	6	lab-04/performance-smoke.api.test.ts	2
lab-04/requester-dashboard.api.test.ts	9	lab-04/route-inventory.test.ts	2
lab-04/seed.test.ts	13	lab-04/staff-dashboard.api.test.ts	8
lab-04/ticket-transitions.test.ts	31	lab-04/ticket-workflow.api.test.ts	401

Client — 31 files, 552 tests

lab-01/App.test.tsx	3	lab-02/AppShell.test.tsx	1
lab-02/AttachmentSection.test.tsx	82	lab-02/Badges.test.tsx	11
lab-02/CreateTicket.test.tsx	18	lab-02/MyTickets.test.tsx	27
lab-02/RequesterTicketDetail.test.tsx	12	lab-02/formatDateTime.test.ts	7
lab-02/ui-style.test.tsx	24	lab-03/AppShell.test.tsx	4
lab-03/ChangePassword.test.tsx	12	lab-03/Login.test.tsx	10
lab-03/RequireRole.test.tsx	16	lab-03/StaffTicketDetail.test.tsx	34
lab-03/StaffTicketQueue.test.tsx	10	lab-03/TicketDetailComments.test.tsx	16
lab-03/UserBadge.test.tsx	5	lab-03/UserManagement.test.tsx	31
lab-03/ui-style.test.tsx	26	lab-04/ActionsTaken.test.tsx	65
lab-04/ActionsTakenNavigation.test.tsx	1	lab-04/AppShellNav.test.tsx	11
lab-04/DashboardRoute.test.tsx	5	lab-04/DrillDownChips.test.tsx	2
lab-04/ListDrillDown.test.tsx	27	lab-04/RequesterActionsView.test.tsx	9
lab-04/RequesterDashboard.test.tsx	10	lab-04/StaffDashboard.test.tsx	15
lab-04/StaffTicketDetailAdministrator.test.tsx	2	lab-04/TicketWorkflow.test.tsx	28
lab-04/ui-style.test.tsx	28		

E2E — 18 spec files, 352 tests

lab-02/harness.smoke.spec.ts	2	lab-02/requester-ticket-flow.spec.ts	5
lab-02/responsive.spec.ts	52	lab-02/submission-evidence.spec.ts	16
lab-03/accessibility.spec.ts	11	lab-03/authentication.spec.ts	4
lab-03/responsive.spec.ts	88	lab-03/staff-ticket-flow.spec.ts	4
lab-03/submission-evidence.spec.ts	24	lab-03/user-administration.spec.ts	4
lab-04/accessibility.spec.ts	18	lab-04/actions-taken-flow.spec.ts	9
lab-04/dashboards.spec.ts	22	lab-04/drilldown-chips.spec.ts	1
lab-04/regression-smoke.spec.ts	6	lab-04/responsive.spec.ts	63
lab-04/screenshots.spec.ts	18	lab-04/ticket-resolution.spec.ts	5

Passing test output in the terminal. The three commands, as run, end with the summary lines above; the same runs wrote the JSON this table is built from. The visual checklist V-01…V-17 is reproduced in Answer Part 9.

Answer Part 4

AI use with reflection

Link: `docs/lab-04/ai-use.md`.

LLM used. Claude (Anthropic) through Claude Code. The session logs for 12 Oct – 2 Nov show `claude-opus-5-5` (from 12 Oct) and `claude-sonnet-5-5` (from 22 Oct) as the orchestrator that held the contract, split each Issue into small slices, re-ran and verified every slice and drove Git; `claude-sonnet-5-5` subagents as the implementers (one narrow slice each, never allowed to push, open a pull request or merge); and `claude-haiku-5-5` for mechanical test-file touch-ups. The subagent logs before 22 Oct are not on this machine, so the worker model for the earliest slices is not shown there. The human engineer (`N0TAW00D`) read every result, decided every judgement call and answered every review; every merge was done by the reviewer `Palapluem`.

Selected key prompts. Nine prompts copied from my Claude Code sessions, typos left as typed (Thai ones glossed in square brackets; the full table is in `ai-use.md`):

When	My prompt	What came out of it
12 Oct · #92	“pls pre setup my lab04 to following this sheet ‘…/SE+Lab+4.pdf’ comprehensively and hardly following the requirement.”	Issues #92–#100 with the `lab-4` label and a dependency graph, `lab4-staging` cut after Lab 3’s release, and the `docs/lab-04/` scaffold with every required file and test path.
12 Oct · #101	“next review the big one through ‘…/SE+Lab+4.pdf’”	A line-by-line review of the merged specification against the handout; it found a numeric-ordering fix had missed the merge, which became #102.
13 Oct · #104	“so fix 104 and re request again”	Three open review points fixed; each new guard was disabled once to prove its test fails; review re-requested.
13 Oct · #104	“ไม่เข้าใจว่าติดอะไร” [I don’t understand what is blocking]	The reviewer’s fourth round turned into a five-row table of what was asked and what I would do, so I could act on it.
17 Oct · #107	“จัดการได้เลย clarify ให้ด้วย” [Go ahead and handle it, and clarify it for me too]	A small follow-up pull request for #105 (hash test, 413 code, spec notes) before the workflow lane was rebased.
30 Oct · #98, #96	“pls check the exisiting work in task list and PRs list, and delegate work to sub agent sonnet worker.”	The board was synced and the remaining Issues were handed to Sonnet workers, each in its own worktree.
30 Oct · #98, #96	“u can delegate more than 1 sub agent”	Three Sonnet workers ran at the same time, each with its own worktree and test database, none allowed to push.
2 Nov · #112	“dud u set timeout?”	A direct answer: no timeout had been changed; the failing hook was vitest’s 10 s default. That ruled out “raise the timeout” and pointed at the slow fixture.
2 Nov · #112	“make it cheaper”	Fixture users created once per file: 627 s → 102 s for that file, suite 1363 s → 512 s, 1175/1175 passing.

My Reflection

Specification agent. Writing the four contract documents before any code again helped most. When a question came up (may a Resolve happen with an open Action, what does a stale update return, in which order are checks made) the answer was already written, so the coding agent did not guess. The contract still needed correcting twice. The reviewer of #101 found that the strict resolution gate would break two earlier tests while the plan said

it listed every edit, and I kept the gate and recorded the consequence (D-27) instead of loosening the rule. On #107 the spec said text is trimmed while the code stores `attachmentNotes` verbatim, and I changed the wording to state the exception, not the code. I think the agent should not decide to change the contract it is building against; I decided each of those.

Coding agent. I split each Issue into phases of small tasks so I could read every result, and I did not accept "all tests pass" from an agent: I re-ran the suites myself and disabled a guard on purpose to check that the right test failed. That caught real gaps, for example an attachment guard that no test actually proved. Review still mattered on top of it: the reviewer found reseed and validation-order bugs that my checks had missed (#104, #105), and a missing test-first order in #110 that I could only record as a deviation because pushed history is not rewritten. A last-week lesson was about test cost: the slowest file made the full suite time out under load, and the answer was to make the fixture cheaper, not to widen the timeout.

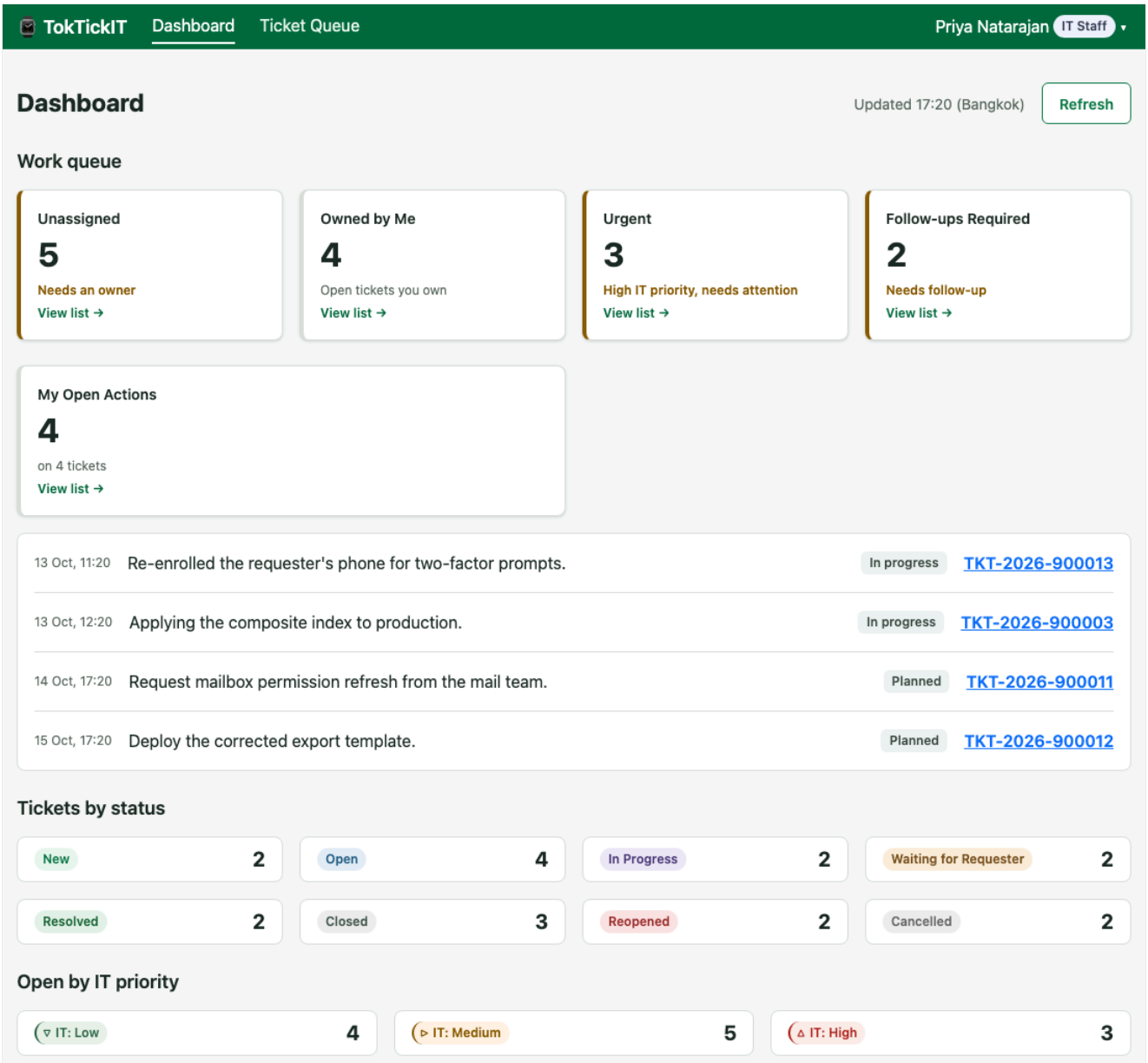
Next time. I would put the standing rules into the first brief (run tests in the foreground, one small task per agent, no co-author line) instead of adding them after something broke, and I would have the agent check for leftover processes on a shared port before a long run: the agent left a stale server behind, so one end-to-end run never started while I waited for it. I would also write test fixtures to create users once per file from the start.

Answer Part 5

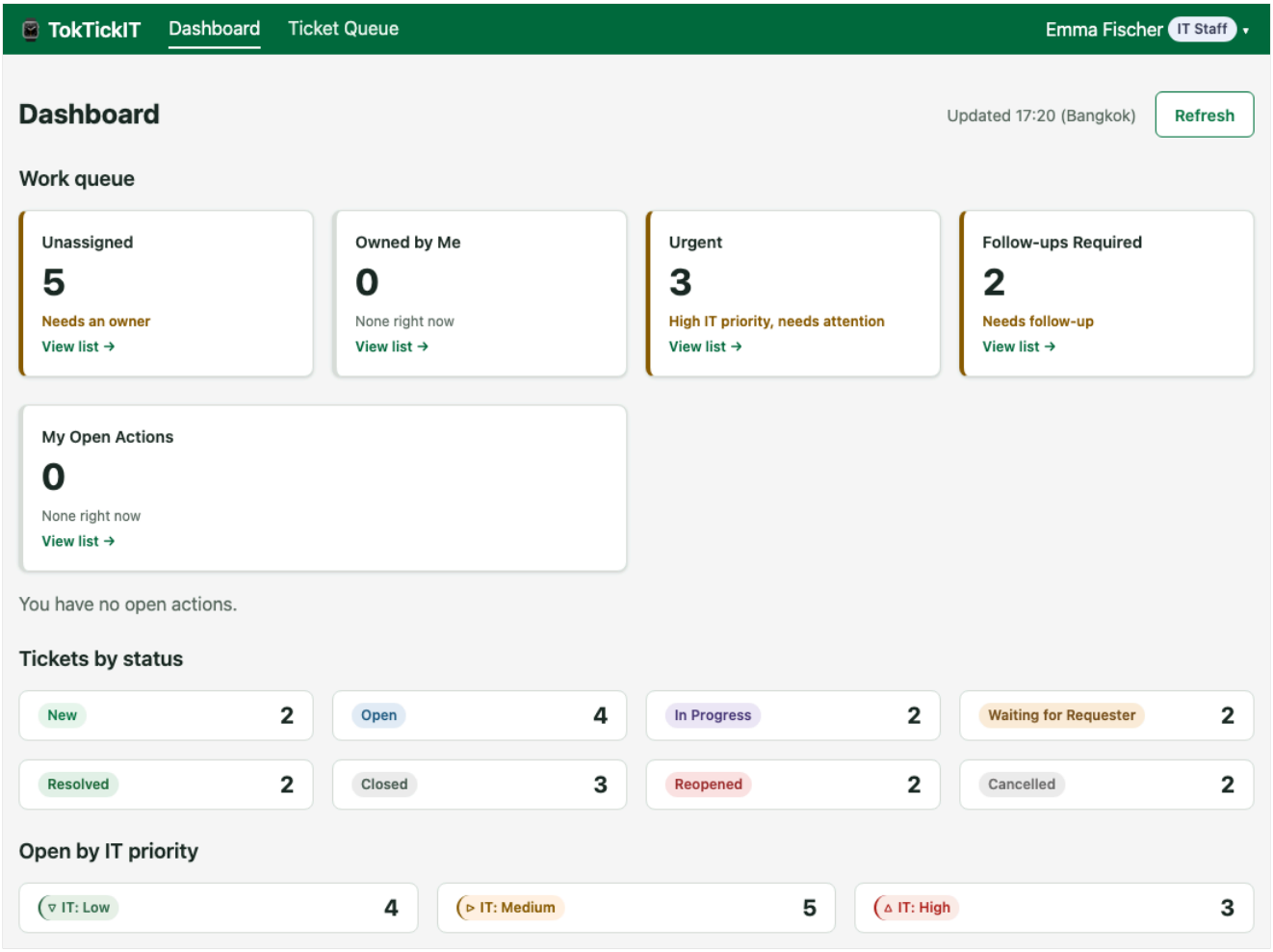
Working IT Staff Dashboard UI

Evidence: `artifacts/lab-04/screenshots/staff-dashboard/` (Playwright spec `e2e/lab-04/screenshots.spec.ts`, E2E-19) and the tests named in each caption.

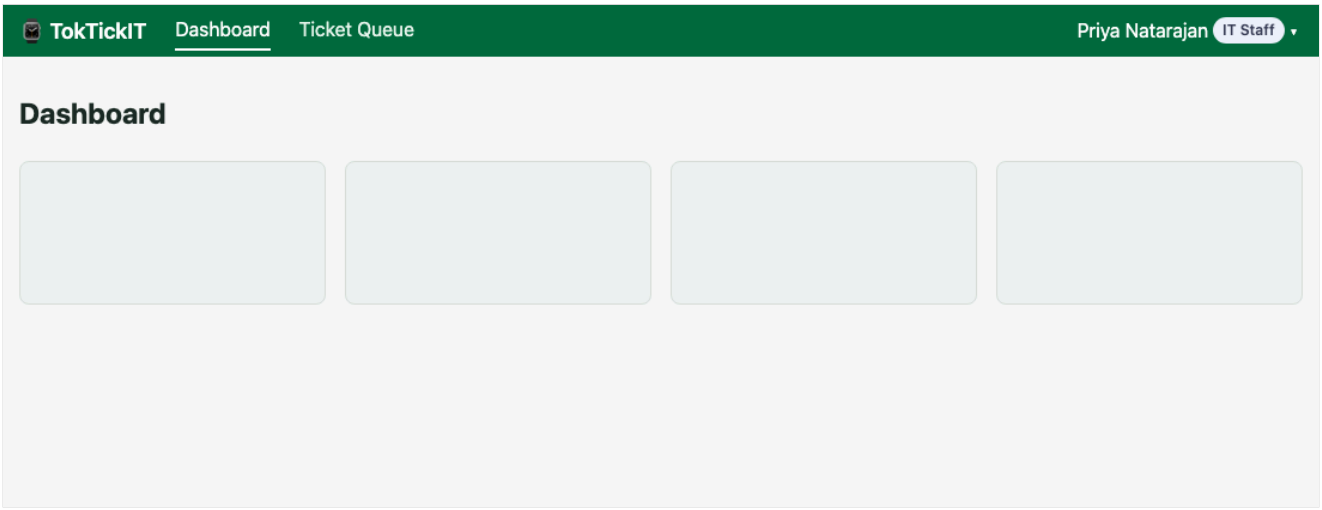
Approved operational metrics, current-user Actions Taken, recent and urgent Tickets, accurate counts. The IT Staff dashboard shows the metrics `ui-spec.md` §3 and `api-spec.md` §5 define: Unassigned, Owned by me, Urgent, Follow-ups, counts by status and by IT Priority, My Open Actions (the current user's own Actions Taken), and Recently Updated Tickets.



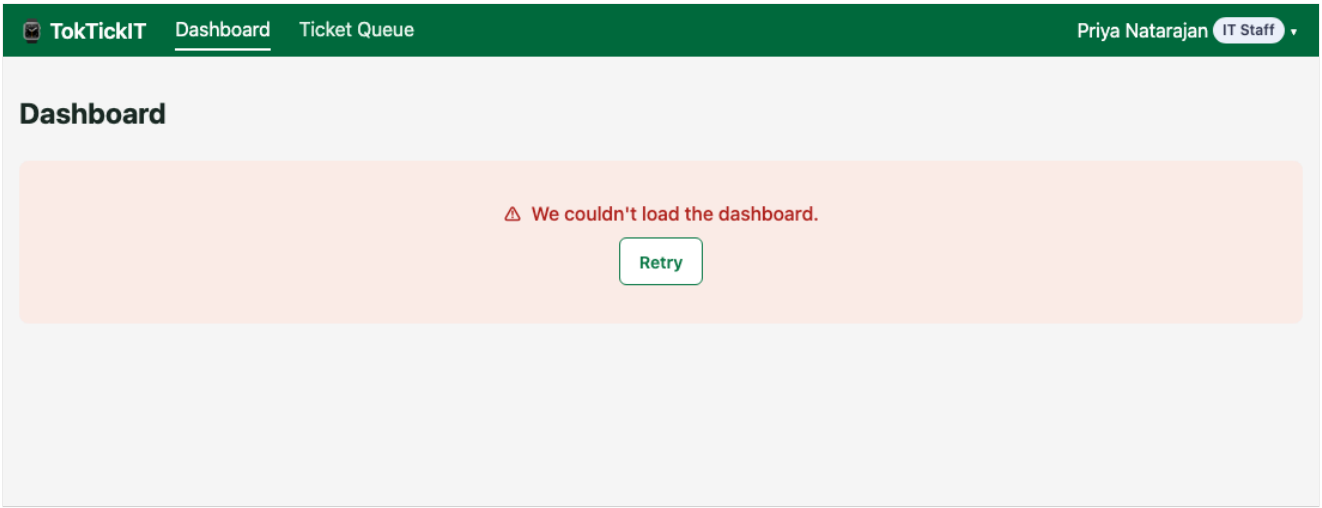
1 — **Loaded** IT Staff dashboard at desktop: metric cards, status and IT Priority counts, the current user's **My Open Actions**, Urgent and Recently Updated lists, each with a drill-down link (C-03, C-04).



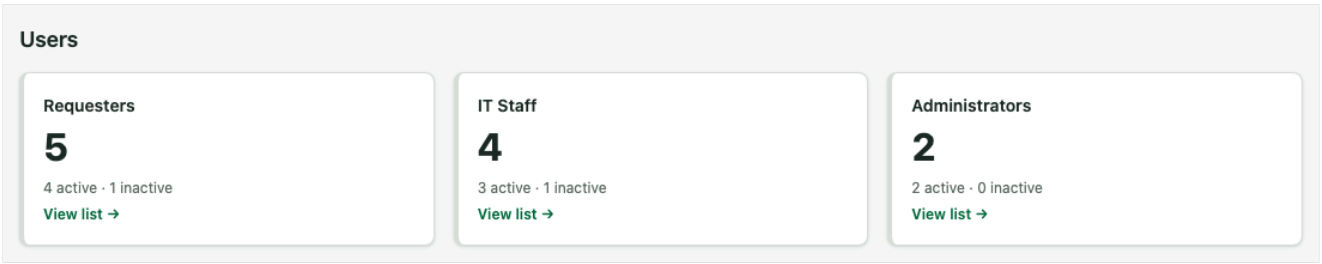
2 — **Zero state**: a staff member who owns nothing and has no open Actions sees **Owned by Me 0** and **My Open Actions 0** with “You have no open actions.”, while the team-wide counts stay correct and nothing errors (API-27, E2E-14, C-06).



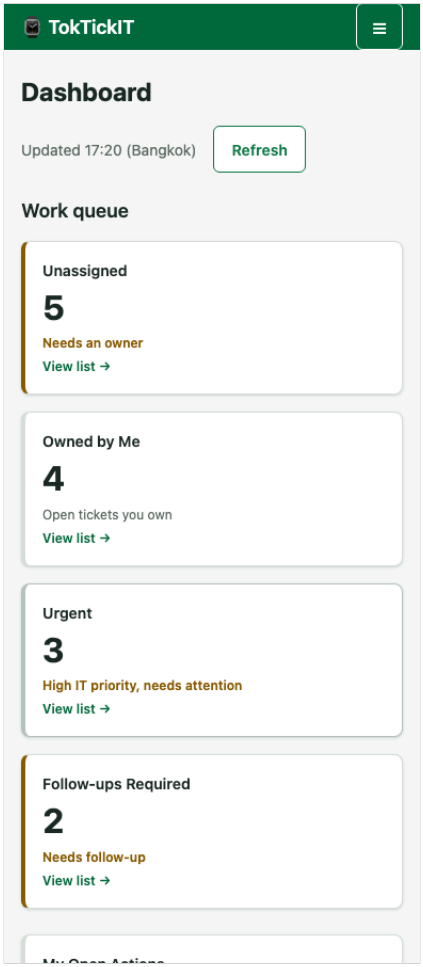
3 — **Loading** feedback while the dashboard is fetched (C-06).



4 — **Safe failure**: an intercepted `500` shows a failure message with Retry and no console error (E2E-16).



5 — **Administrator view** (bottom of the same dashboard): the Users strip with active and inactive counts per role, each with a View list link to `/admin/users?role=…`; IT Staff users do not get it (API-26, API-31, E2E-13).



6 — **Responsive behavior**: the same dashboard at 390 px (top of the page; no horizontal scroll, R-01).

Drill-down behavior and forbidden state. Each card opens the Ticket Queue with the matching filter and the queue total equals the card value (API-28, E2E-11); a Requester who opens a staff-only URL sees the forbidden state and no protected request is sent (E2E-15).

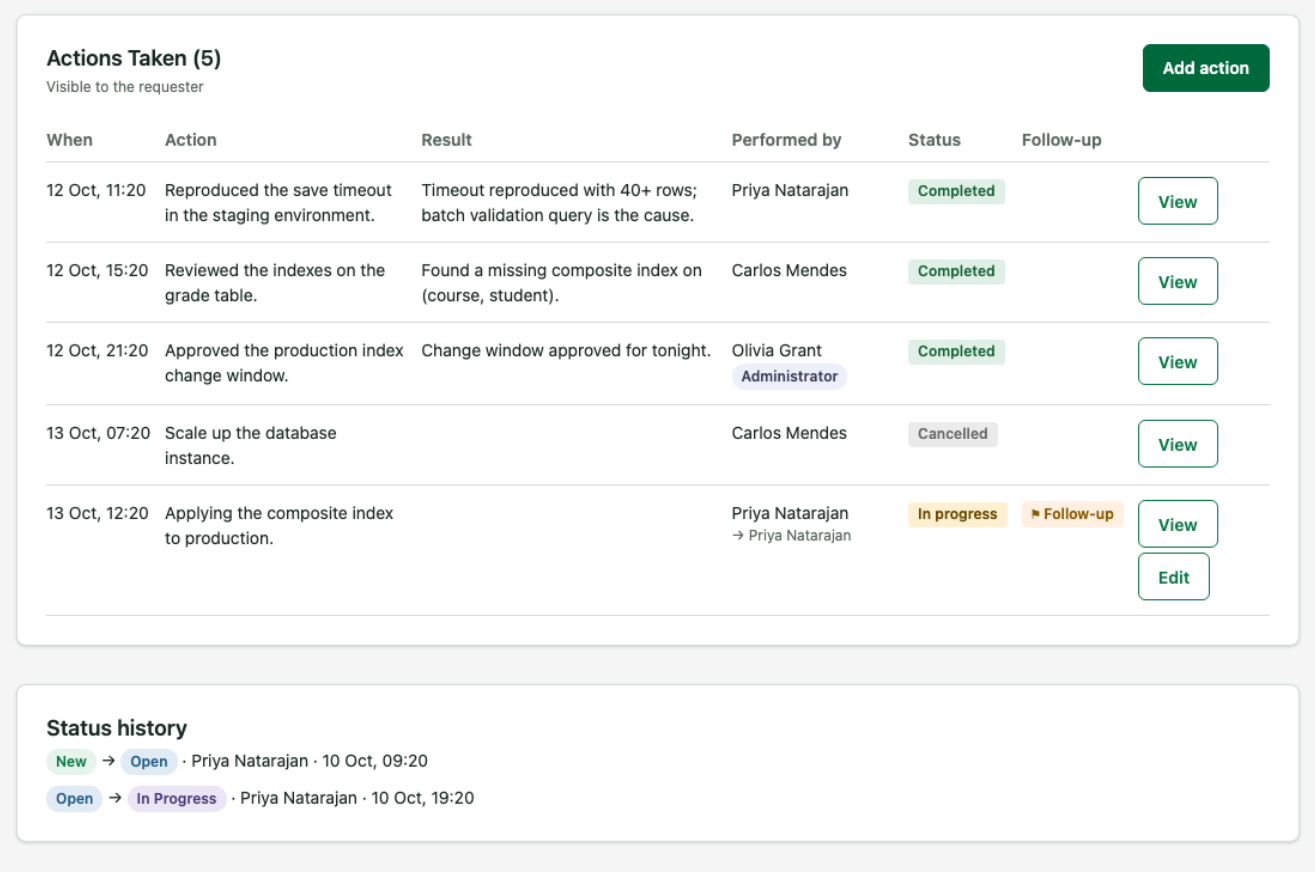
Selected metrics match database queries. The staff metrics are asserted against independent SQL over the seeded database, not against the code under test:

Row	What the test compares to the database	Final
API-23	`unassigned`, `ownedByMe`, `byStatus` (all eight keys) and `byItPriority` equal independent SQL	Pass
API-24	My Open Actions counts Planned/In Progress Actions on open Tickets for assignee = me (or none and performer = me)	Pass
API-25	Follow-ups, Urgent and Recently Updated equal independent SQL	Pass
API-28	Every card's drill-down queue total equals the card value	Pass
API-31	Each user-count role `active + inactive` equals the length of `GET /api/users?role=<ROLE>`	Pass

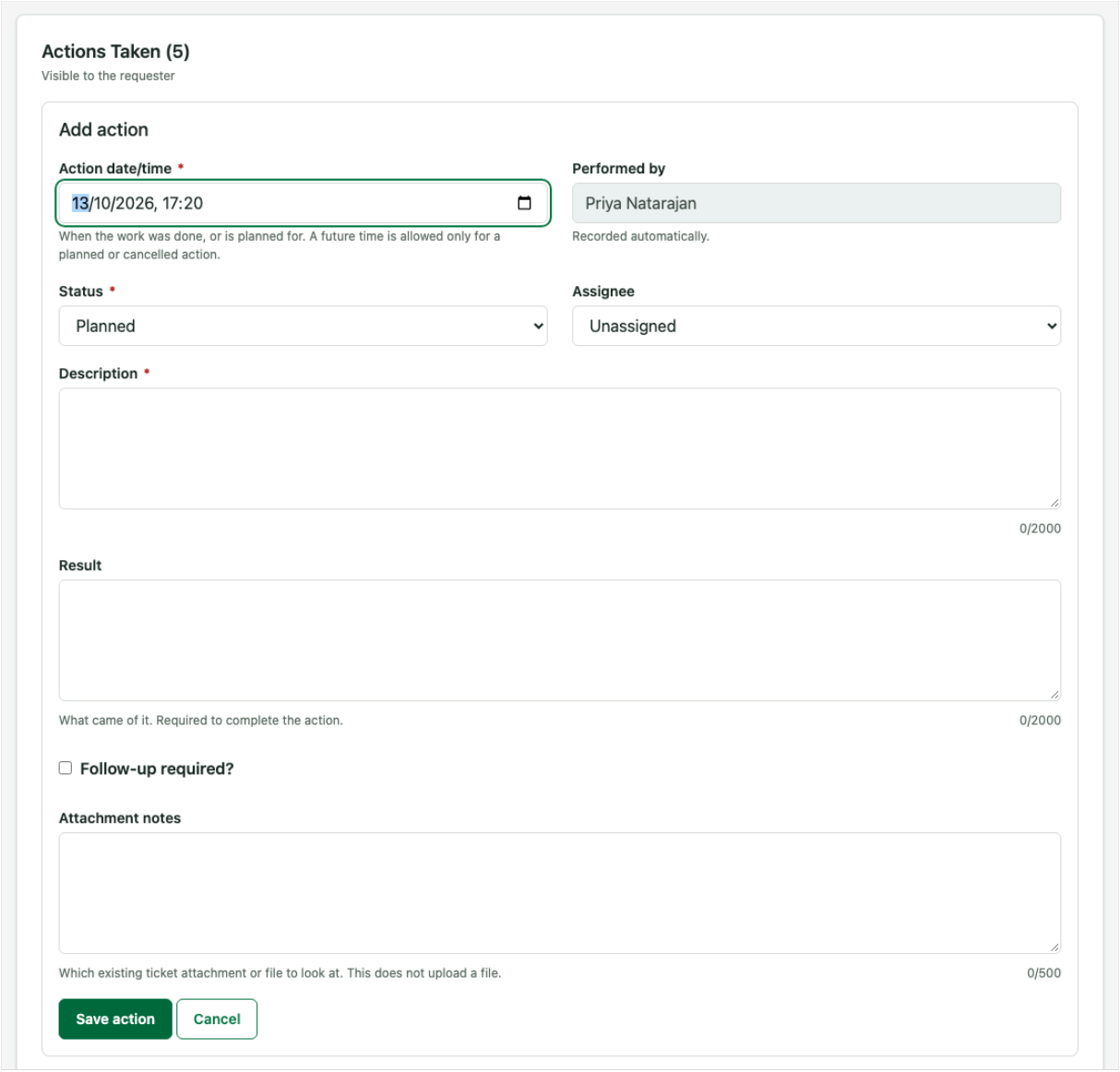
Answer Part 6

Working Actions Taken UI

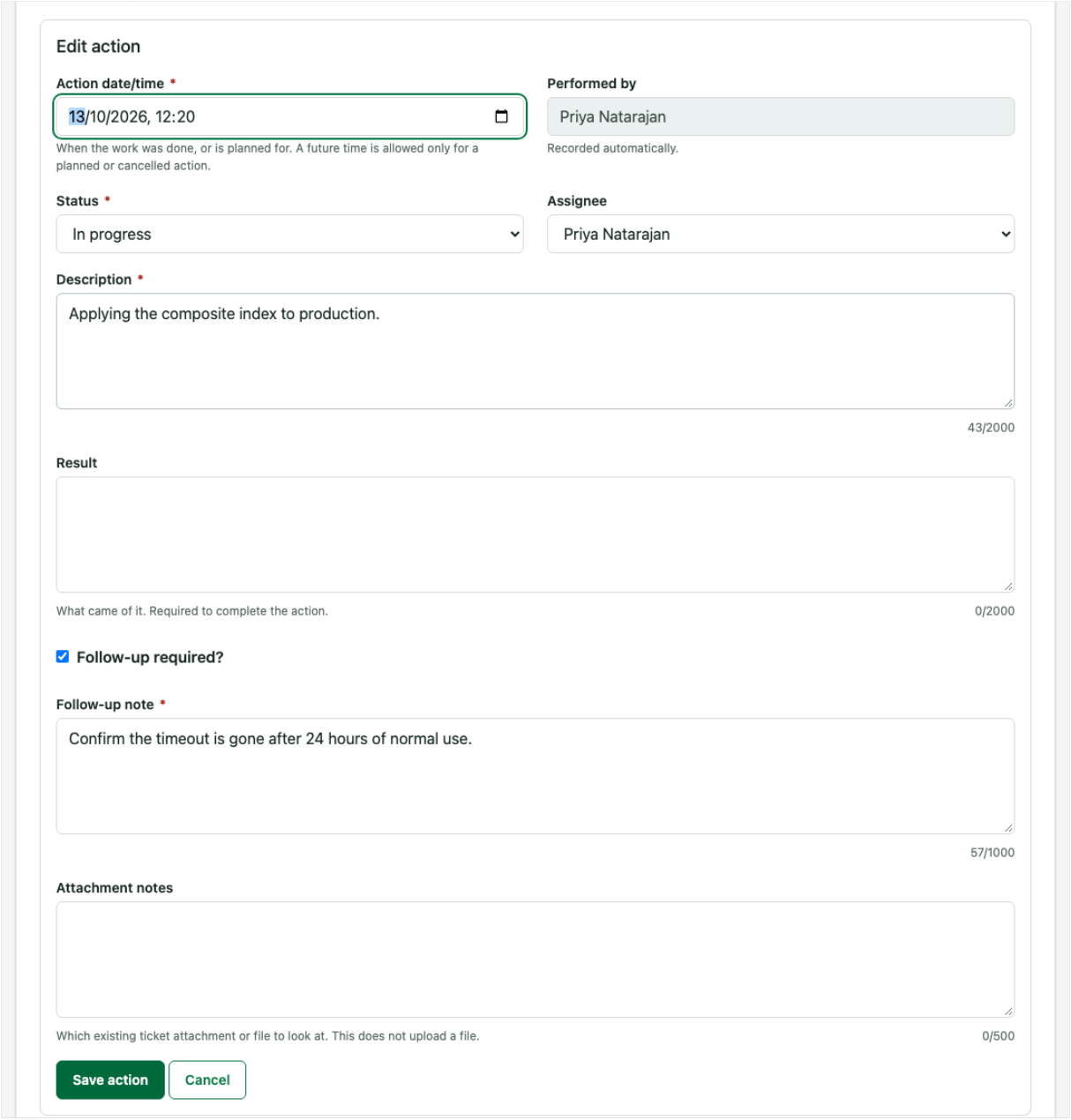
Evidence: `artifacts/lab-04/screenshots/actions-taken/` (E2E-19) and `e2e/lab-04/actions-taken-flow.spec.ts` (E2E-01..06).



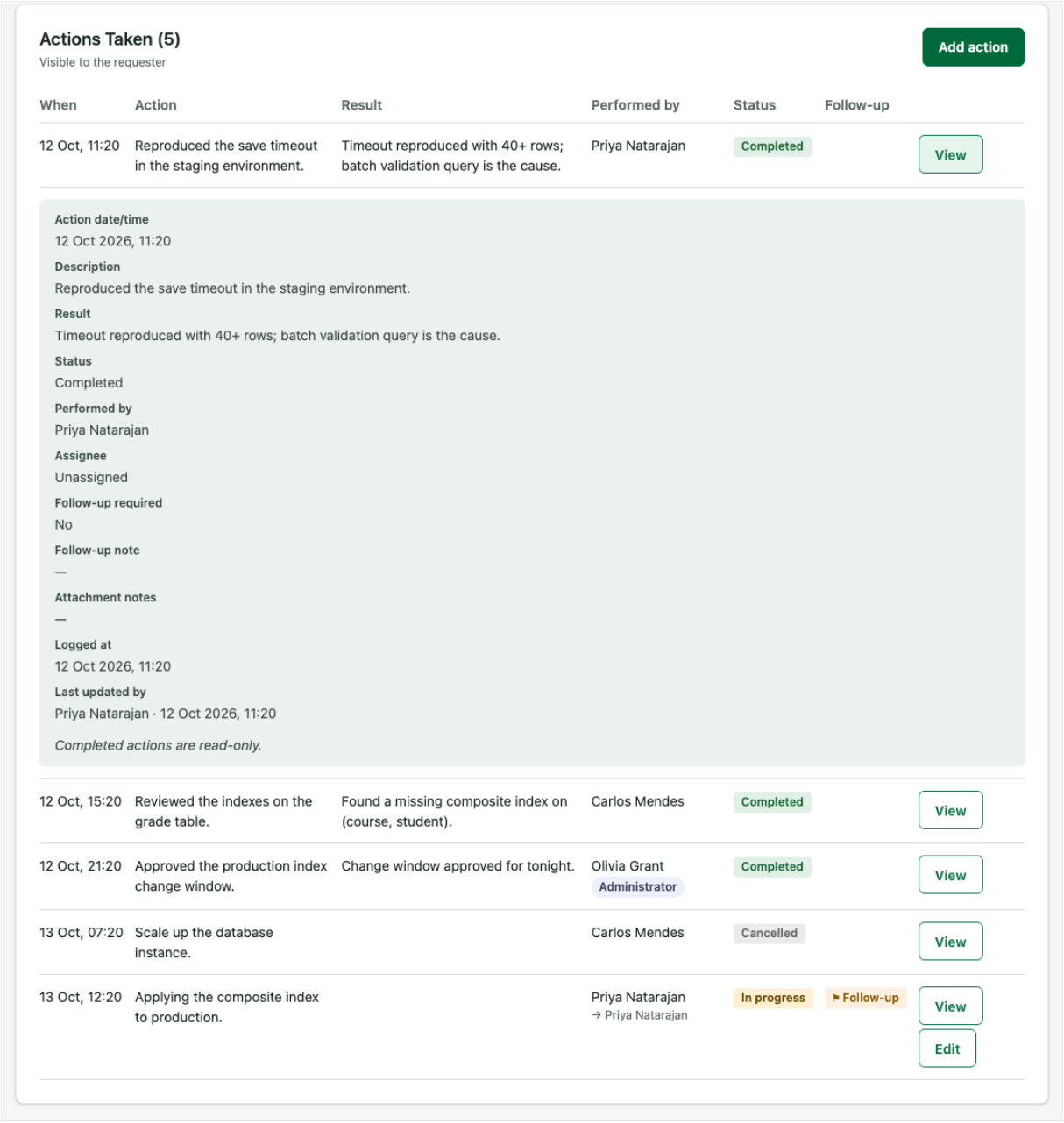
1 — **List**: several different Actions Taken on one Ticket, each with date/time, description, result, performer, assignee, follow-up and status badge; Edit/View per row (C-10, E2E-02).



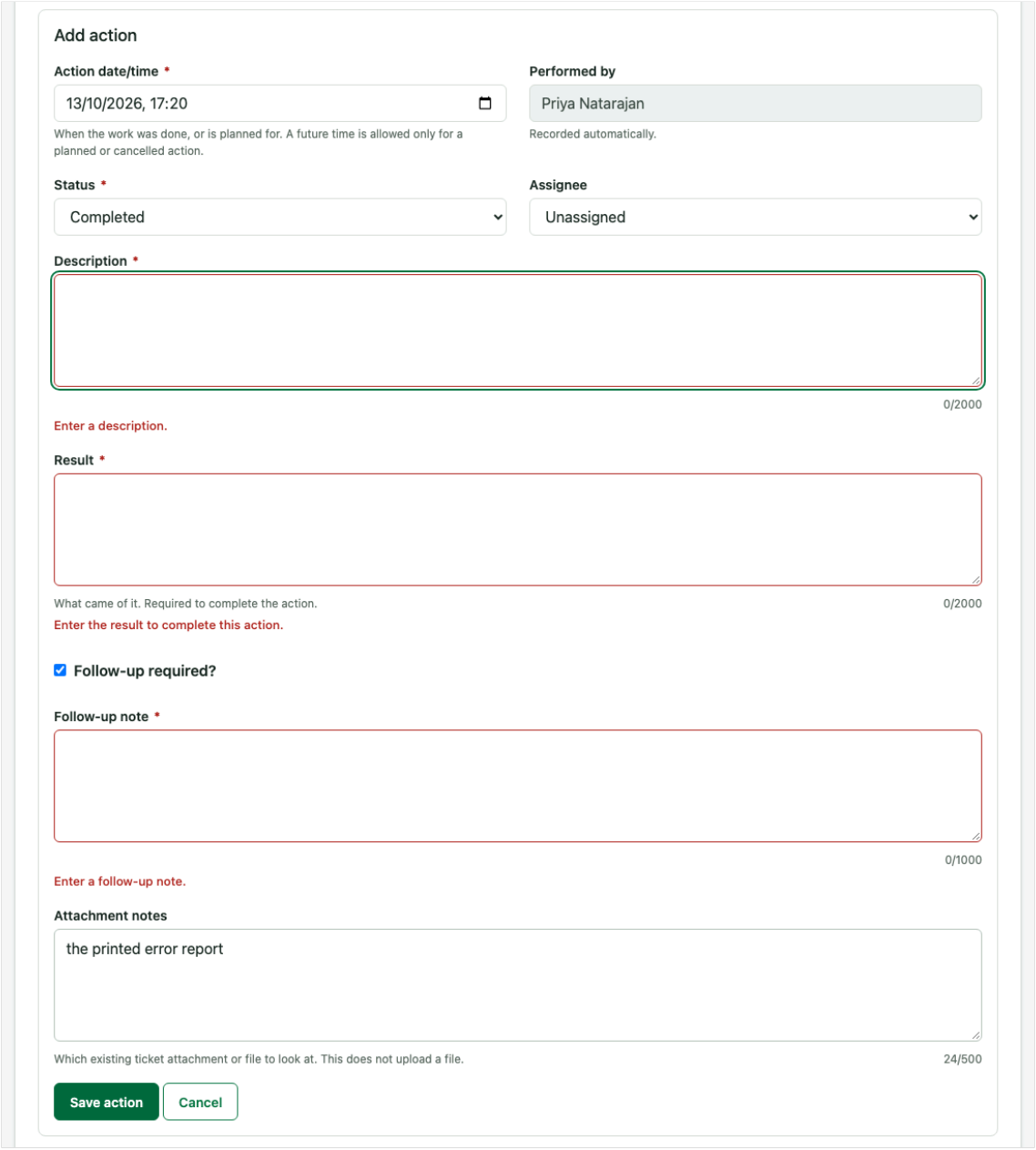
2 — **Create**: the form with date/time, status, assignee (active IT Staff and Administrators only), follow-up with its conditional note, description, result and attachment notes; Performed by is shown, not editable (C-11, C-13).



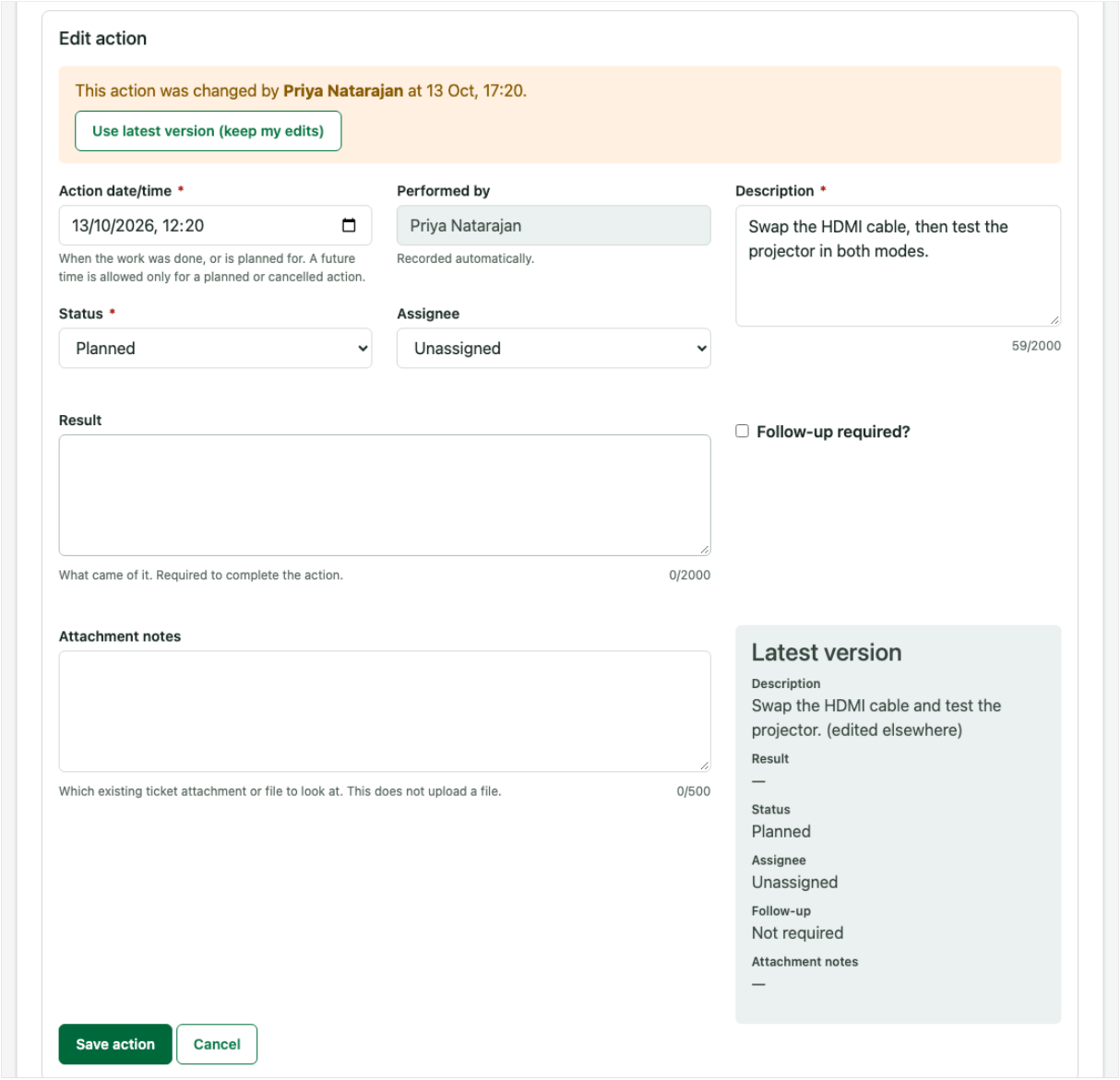
3 — **Edit** an open Action: the form is pre-filled and the status control offers only the permitted next states; Planned → In Progress → Completed, and Cancelled behind a confirm dialog, are exercised by C-18 and E2E-03.



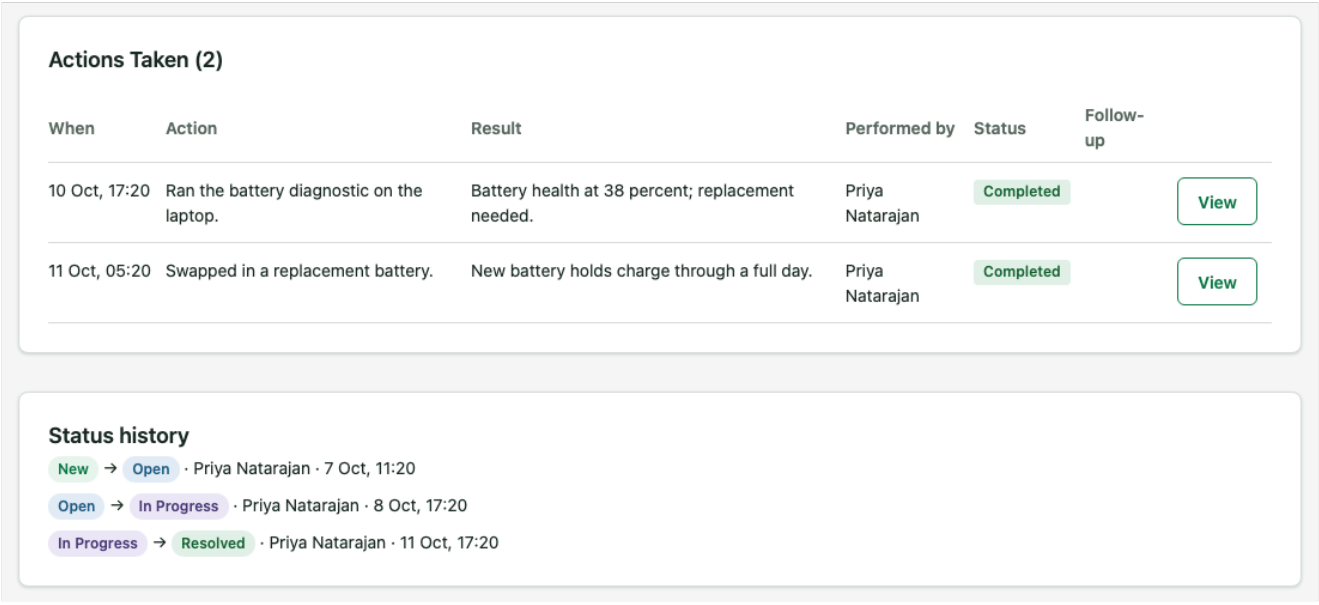
4 — **View mode**: read-only fields on the read-only surface; a Completed or Cancelled Action loses Edit (C-17, S-05).



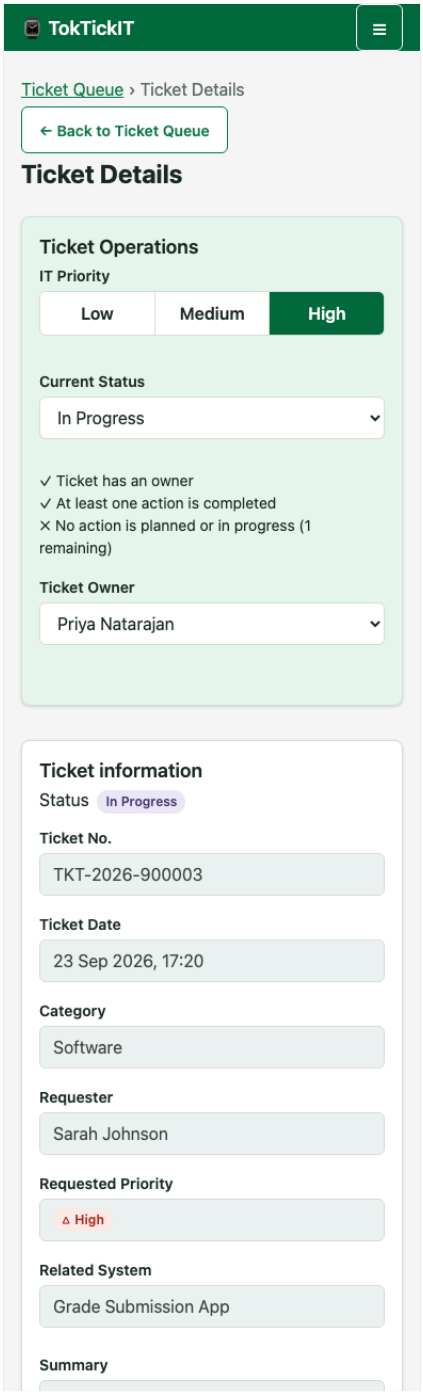
5 — **Validation**: messages sit directly below their field and the typed values are kept (C-12, E2E-04).



6 — **Stale update**: the second editor sees who changed the Action and when, the current values including the date/time, and keeps their typed edit (C-15, E2E-05).



7 — **Role restriction**: the owning Requester sees every Action Taken read-only, with no write controls and no Internal Notes (C-27, E2E-09).



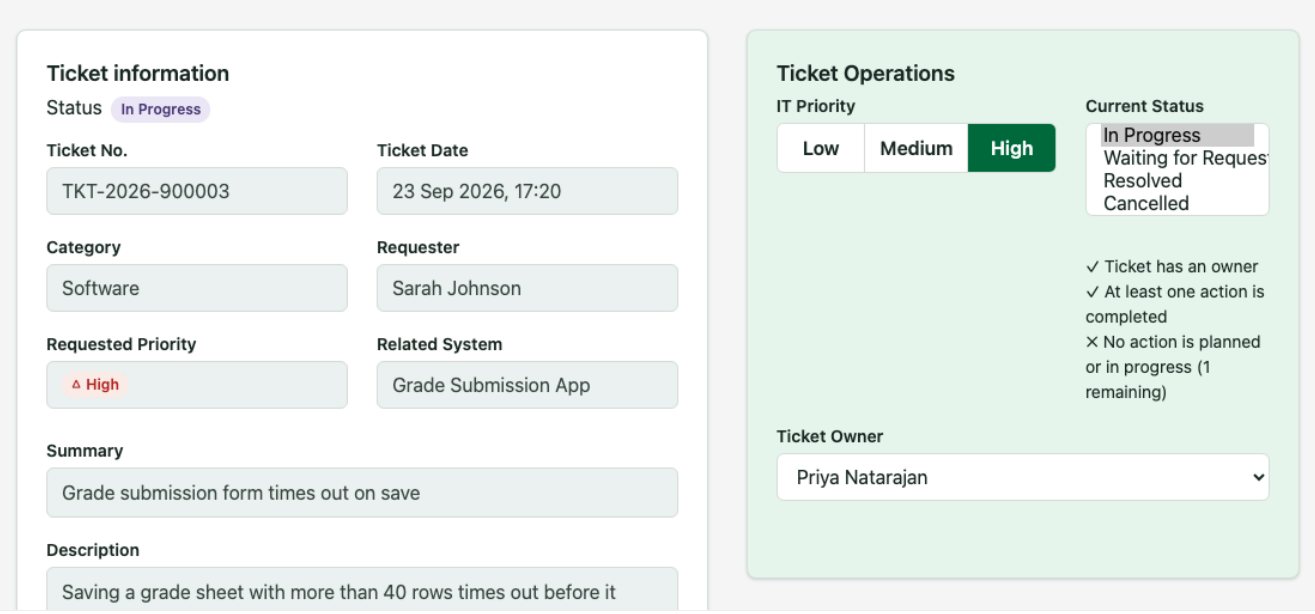
8 — **Responsive behavior**: below 768 px the table becomes stacked cards (top of the page, R-03).

Assign, inactive-assignee rejection, role restrictions and safe failures are proven by tests, not screenshots alone: assigning to an active staff member (API-01, E2E-01), refusing an invalid or deactivated assignee with a field error (API-07, C-19, E2E-04), Requesters refused on every Actions write (SEC-01) while Administrators keep access with their limits (SEC-02), duplicate submit and a dropped-then-retried request creating exactly one Action (C-16, API-13, API-33, E2E-06), and the failure state with Retry for a `500` (C-28, E2E-16). All of these rows are Pass in `tests.md`.

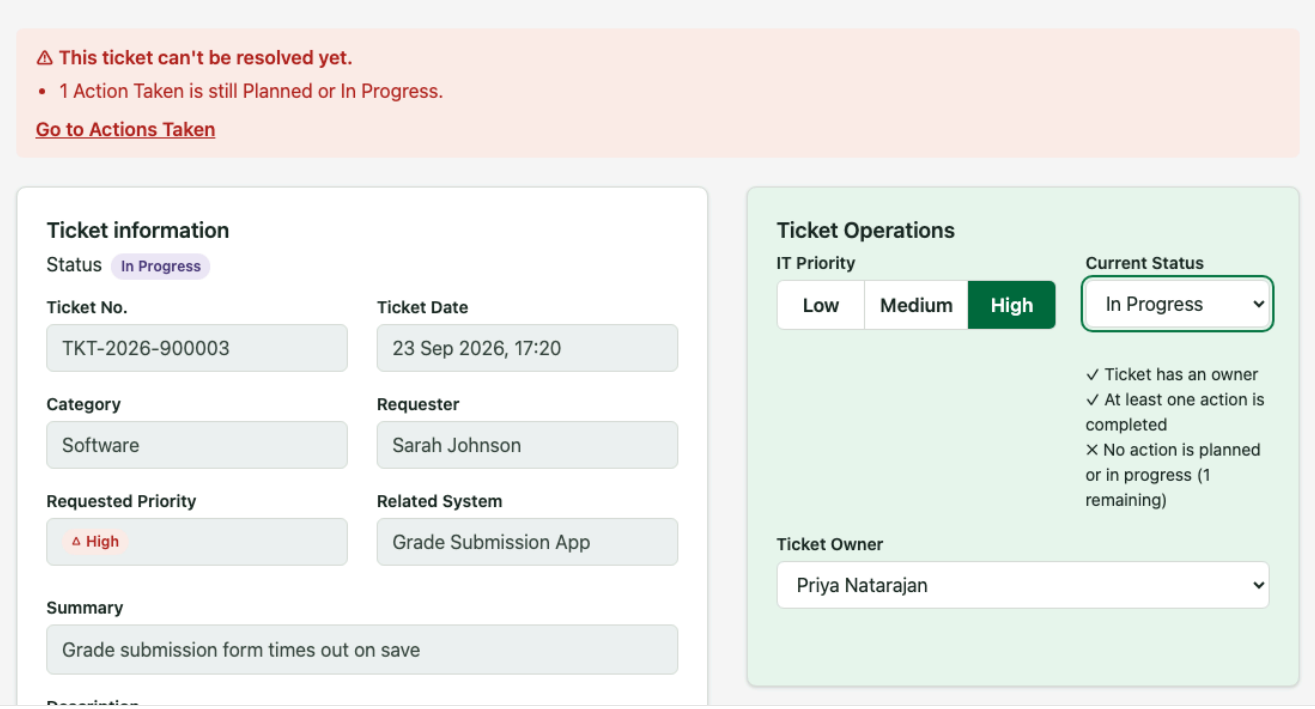
Answer Part 7

Working Ticket Workflow

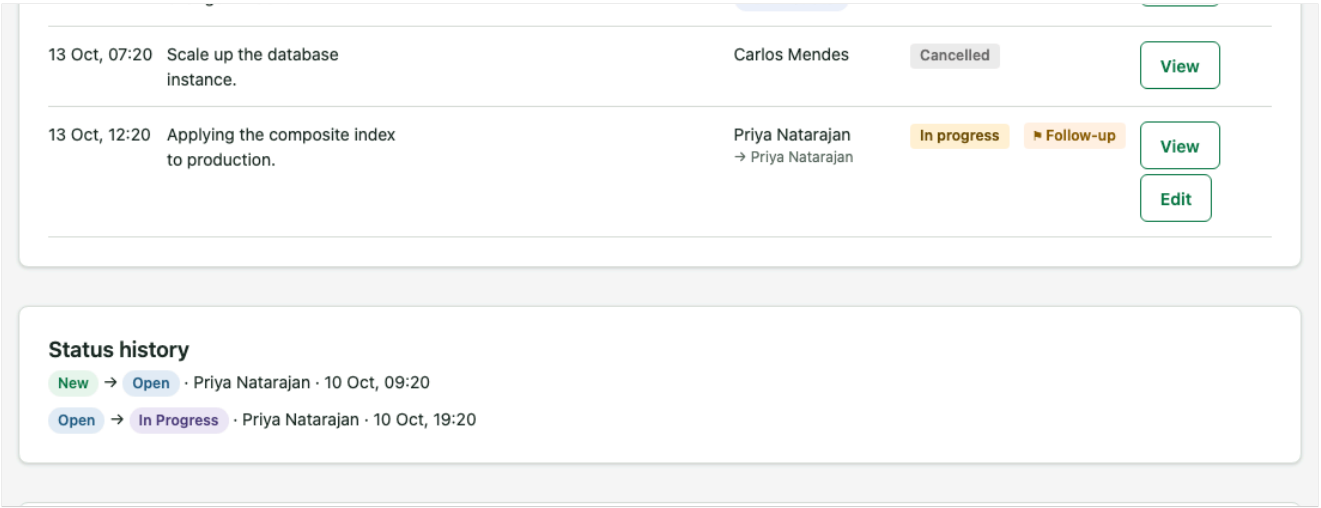
Evidence: `artifacts/lab-04/screenshots/ticket-workflow/` (E2E-19), `e2e/lab-04/ticket-resolution.spec.ts` (E2E-07..10) and `server/tests/lab-04/ticket-workflow.api.test.ts` (WF-01..16).



1 — **Permitted transitions**: the status control offers only the transitions the matrix allows for the Ticket's current status; the resolution checklist matches the Ticket's data (C-22, C-24, WF-01, WF-02).



2 — **Resolution gate**: Resolved is blocked with every failed reason listed; after the open Action is completed the same control succeeds (WF-03..06, E2E-07).



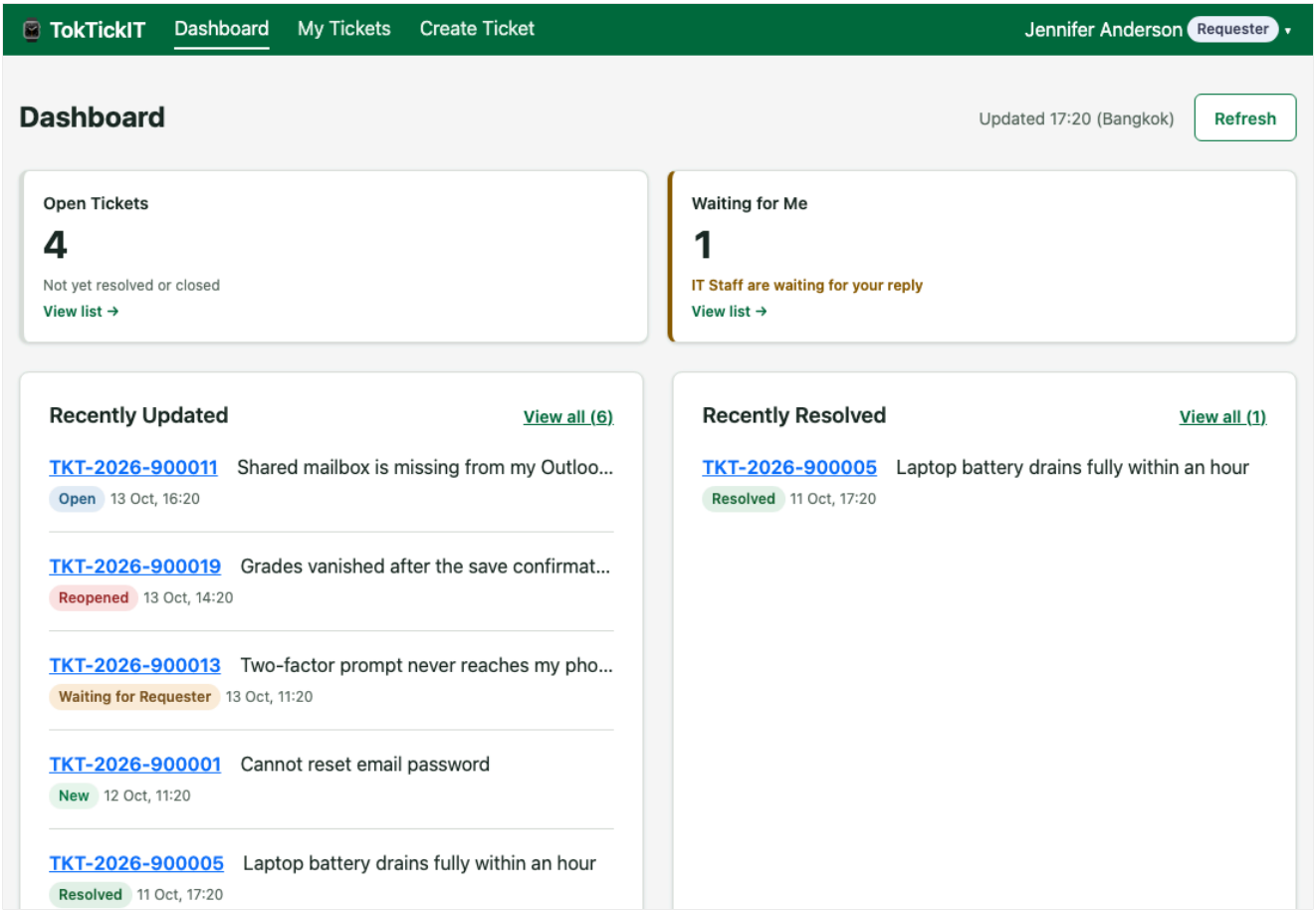
3 — **Append-only history in stable order**: each change with from, to, who and when, ordered by time then id; visible to IT Staff, Administrators and the owning Requester (WF-08, WF-09, E2E-09).

Stable ordering and append-only behaviour are asserted at the HTTP layer: history is returned in `changedAt` then `id` order and is `[]` for a Ticket that never changed (API-16, WF-09); POST, PUT, PATCH and DELETE on the history route are `404` for every role and write nothing (WF-09). **Role-appropriate visibility:** the owning Requester reads their own history, another Requester gets the same `404` as for a nonexistent Ticket, and a request with no session is `401` (WF-09, SEC rows). A status change and its history row commit in one transaction, and a forced failure rolls both back (WF-08); a stale status write returns `409 STALE_UPDATE` with the current Ticket (WF-10, E2E-10).

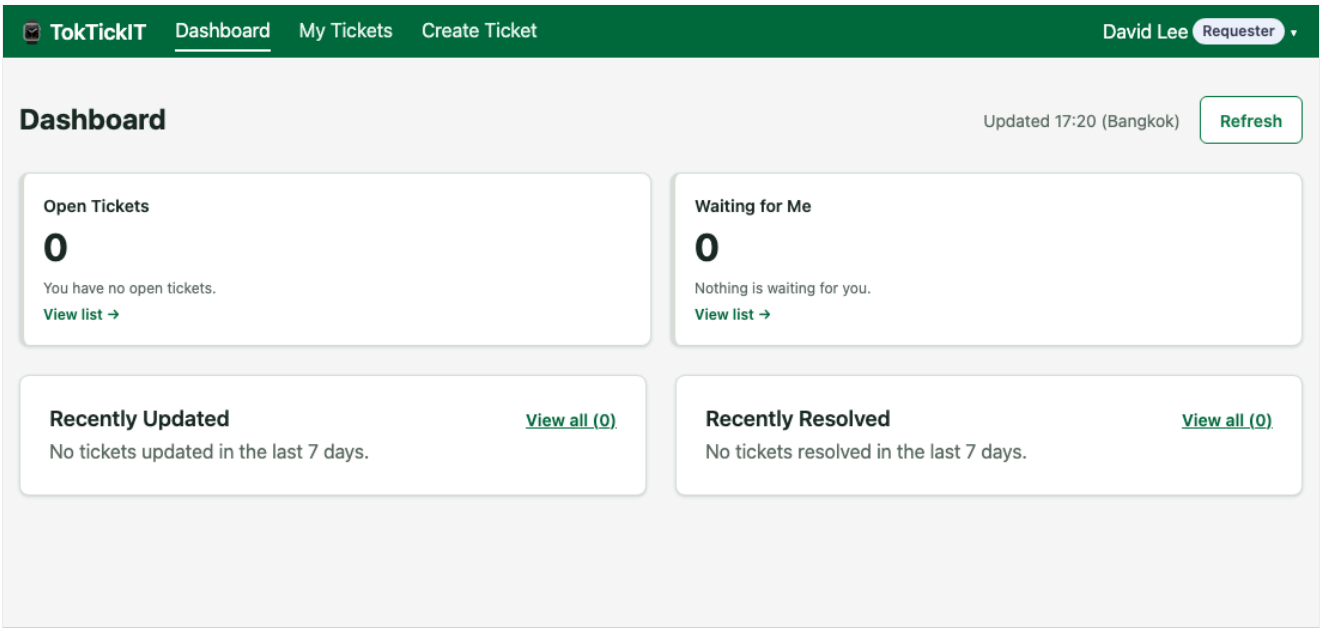
Answer Part 8

Working Requester Dashboard and Final Regression UI

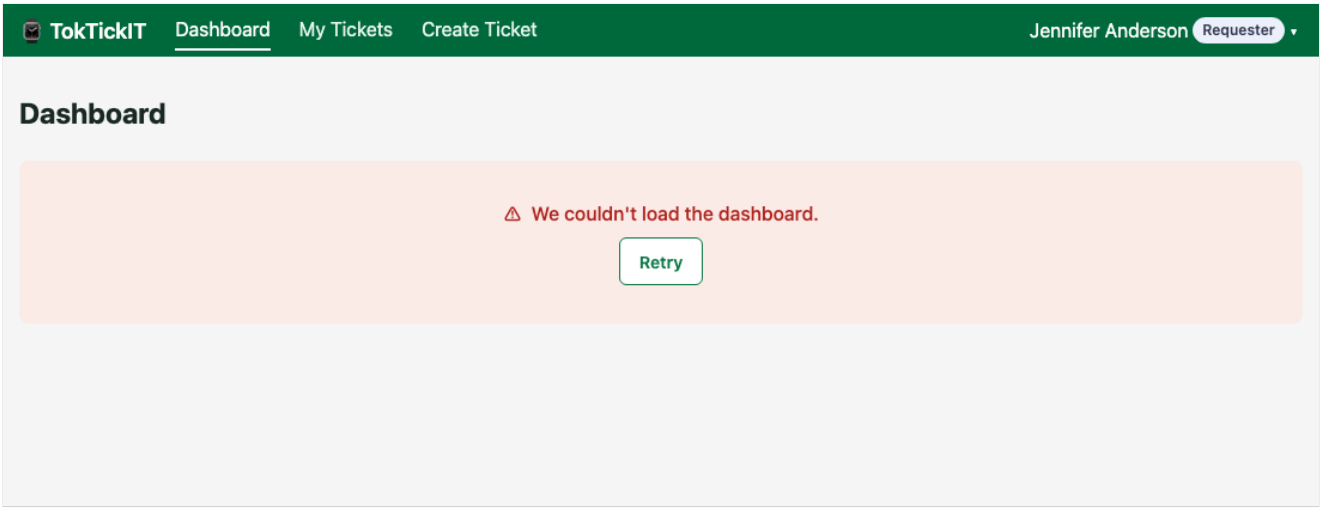
Evidence: `artifacts/lab-04/screenshots/requester-dashboard/` (E2E-19) and `e2e/lab-04/dashboards.spec.ts` (E2E-11..16).



1 — **Requester-owned metrics**: My Open Tickets, Waiting for me, Recently Updated and Recently Resolved for the signed-in Requester only, with recent and attention-required Tickets and “View all” drill-downs (C-08, API-17, API-22).



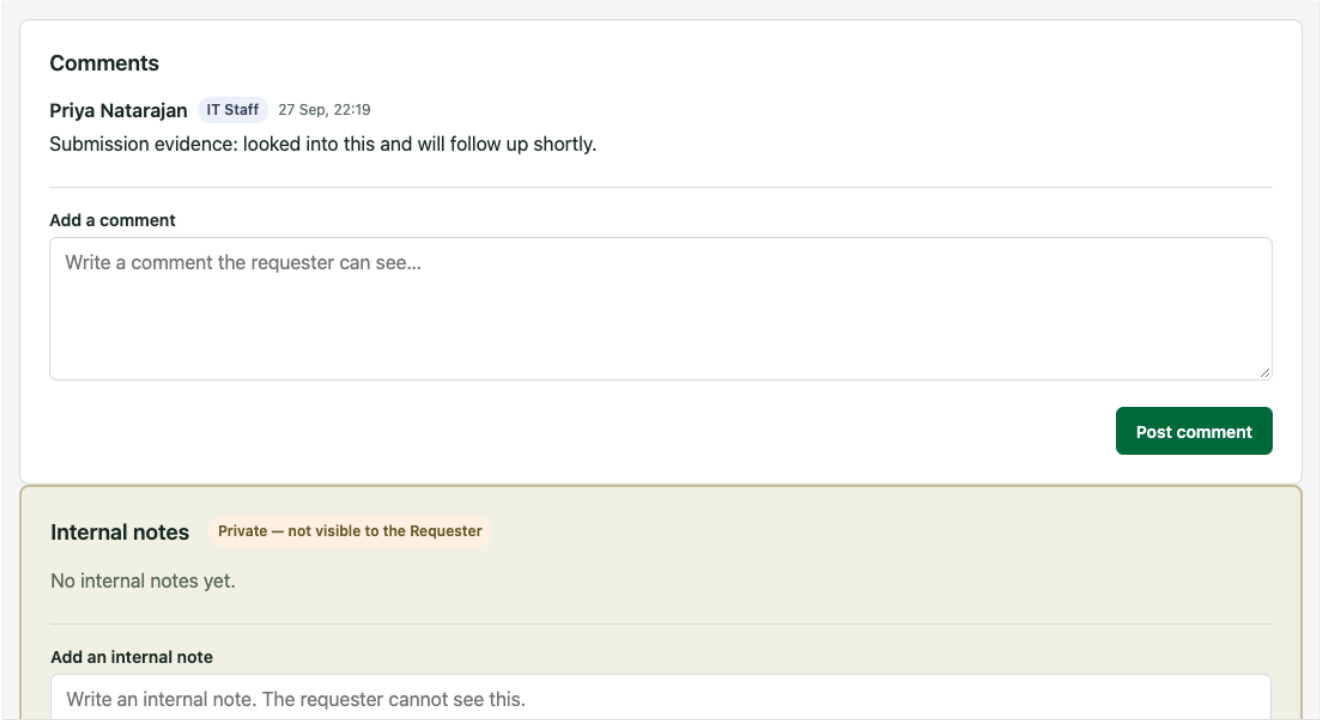
2 — **Zero state**: a Requester with no Tickets sees zero cards and empty messages (API-20, E2E-14, C-09).



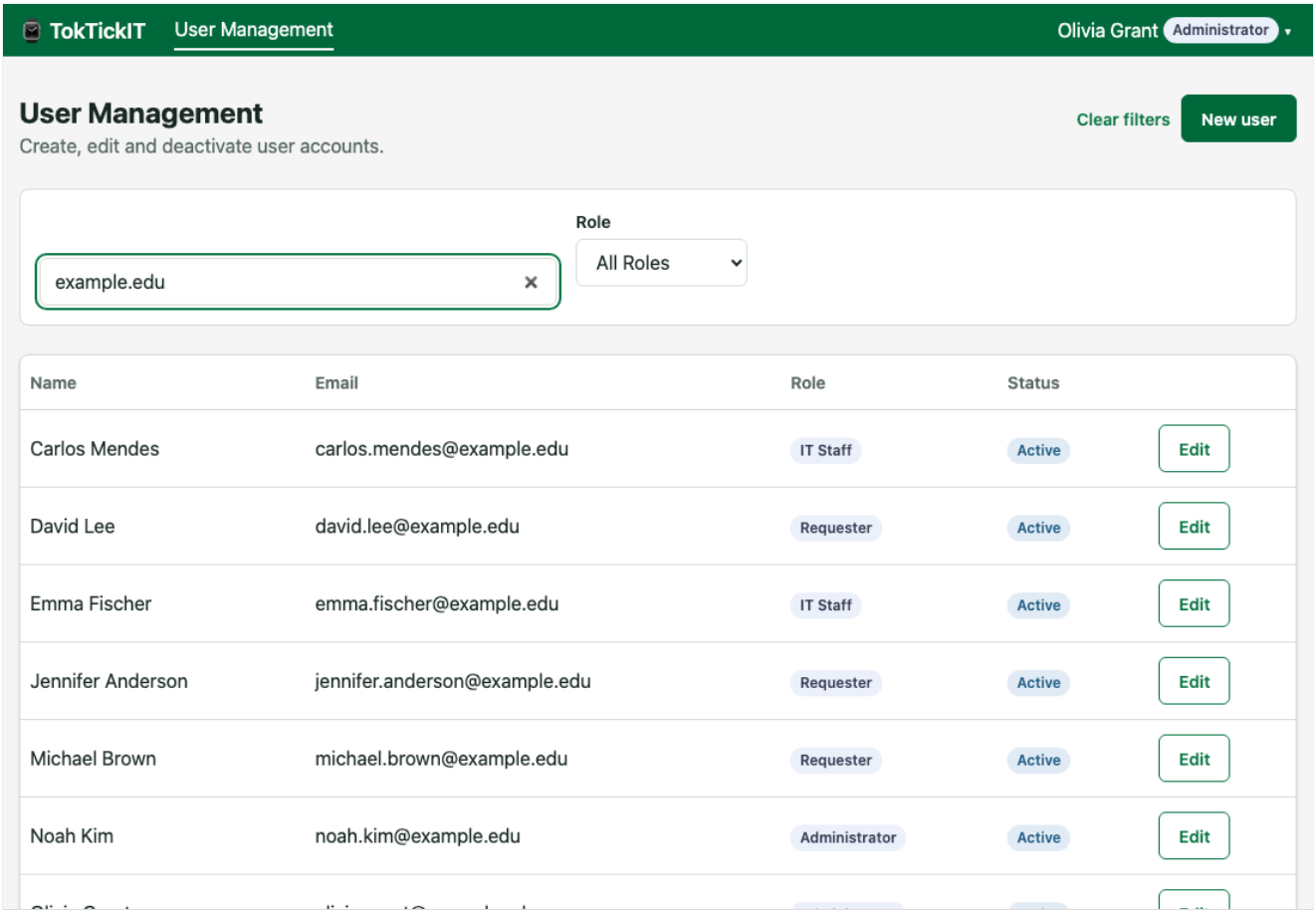
3 — **Safe failure** with Retry (E2E-16).

Ownership protection. Another Requester's Tickets are never counted or listed and a `requesterId` in the query is ignored (API-18); a pasted `/tickets/:id` of another Requester's Ticket is not found (E2E-12, SEC rows).

Representative regression evidence for Labs 1–3. E2E-18 (`e2e/lab-04/regression-smoke.spec.ts`) walks, on the Lab 4 build, login and forced password change, My Tickets, Ticket Detail, attachments, Public Comments, the IT Staff queue and detail, Internal Notes, and Administrator user management; it is Pass. The Labs 1–3 suites themselves are green and unchanged except for the edits recorded in `tests.md` §4.2: server REG-01..03, client REG-04 (349 tests across Labs 1–3) and e2e REG-05..06 (75 Lab 2 and 135 Lab 3 e2e tests) are all Pass. Two of those screens are shown below from the committed Lab 3 screenshot set; E2E-18 and REG-05/06 re-exercise them on the final build.



IT Staff posting a Public Comment on a Ticket (Lab 3 submission set; Lab 4 keeps the thread and adds the Actions Taken and history cards above it)



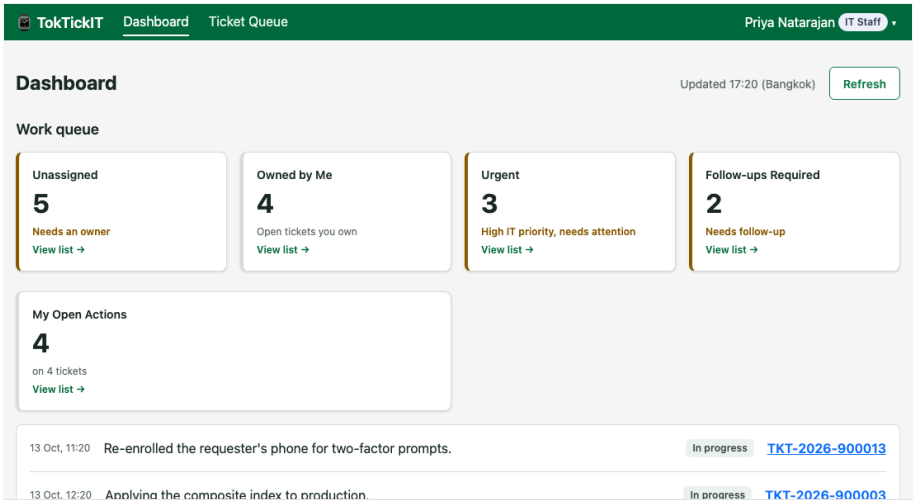
Administrator User Management: list, role and status, edit action (Lab 3 submission set)

Answer Part 9

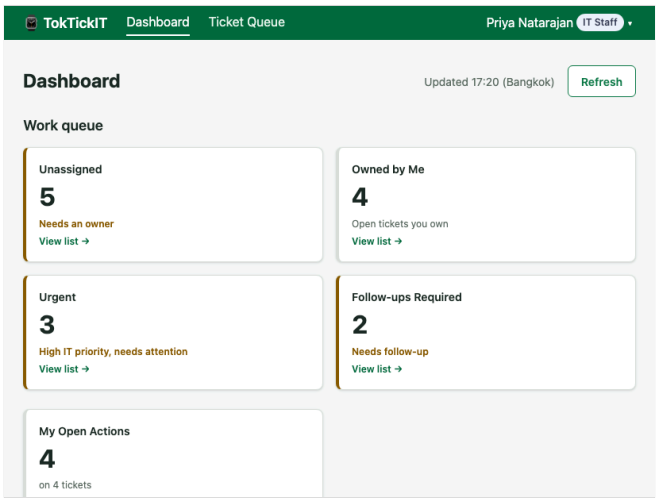
Zen Green UI, responsive, accessibility and final polish

Rendered / linked: `docs/lab-04/ui-spec.md` — Zen Green reuse and extensions (§1), navigation and role landing (§2), IT Staff Dashboard (§3), Requester Dashboard (§4), Actions Taken on Ticket Detail (§5), ticket workflow and resolution feedback (§6), feedback conventions (§7), responsive behavior (§8), accessibility (§9), the visual and accessibility checklist (§10) and screenshot paths (§11).

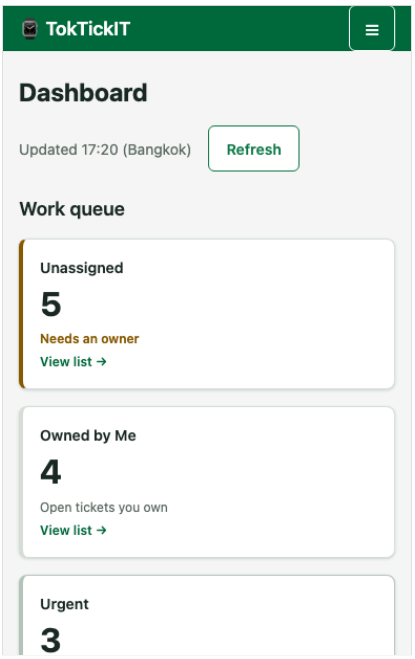
Desktop / tablet / mobile screenshots (`artifacts/lab-04/screenshots/{screen}/{state}-{desktop,tablet,mobile}.png`; 1440×900, 820×1180 and 390×844 viewports; tall pages are cropped to their top here, the full images are committed):



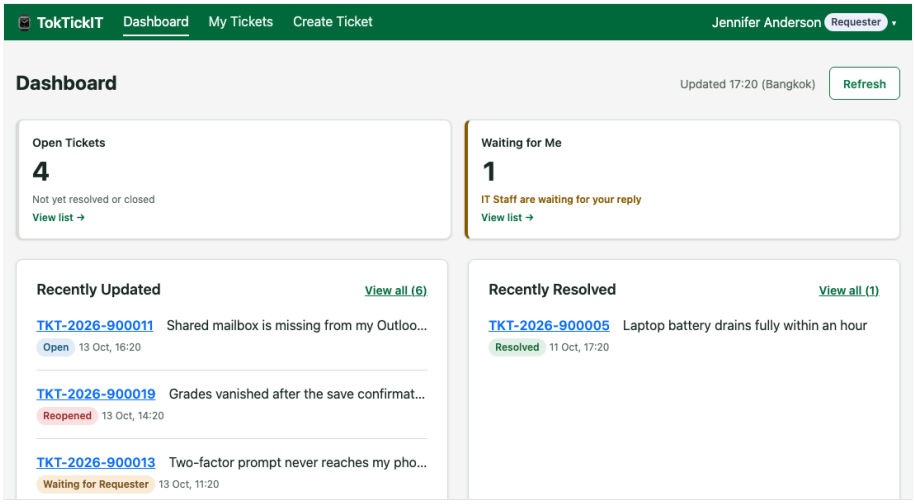
IT Staff Dashboard — desktop



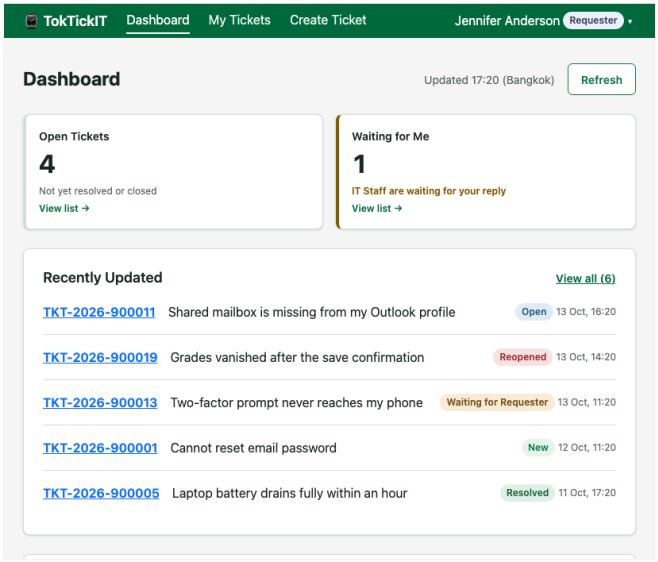
tablet



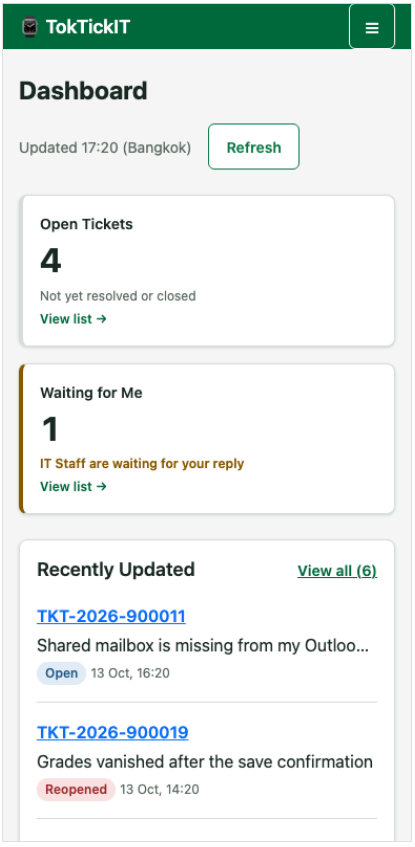
mobile



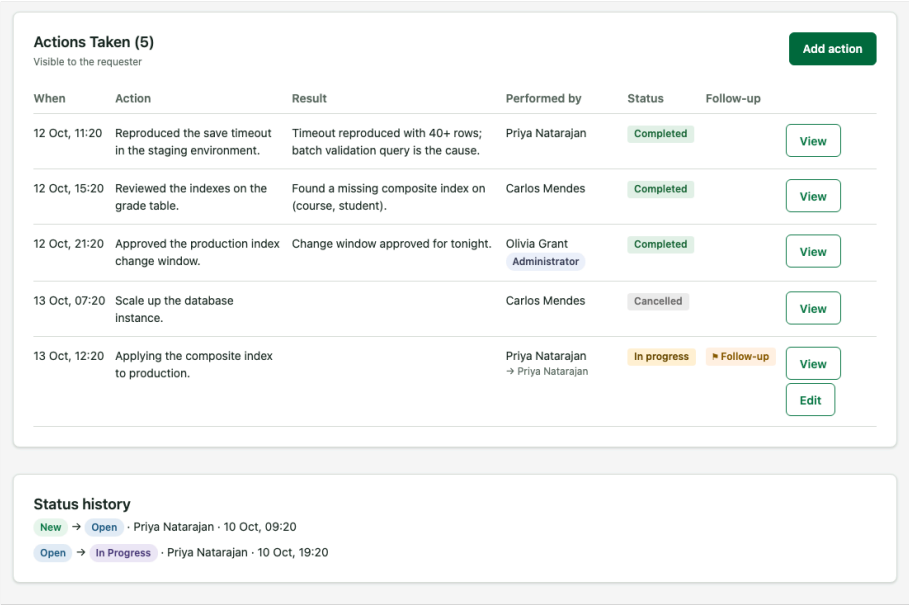
Requester Dashboard — desktop



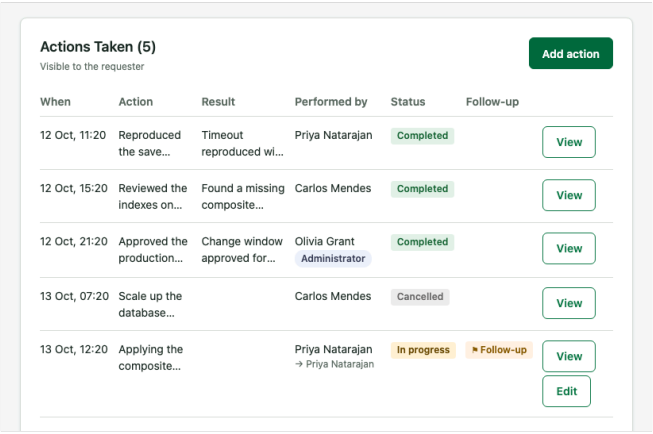
tablet



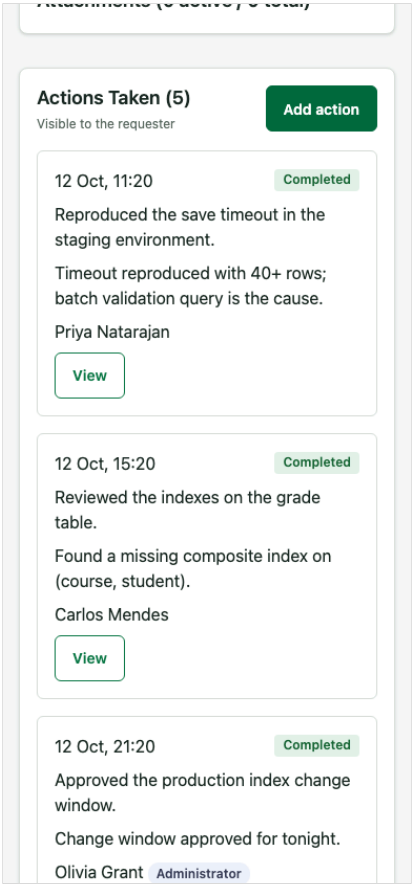
mobile



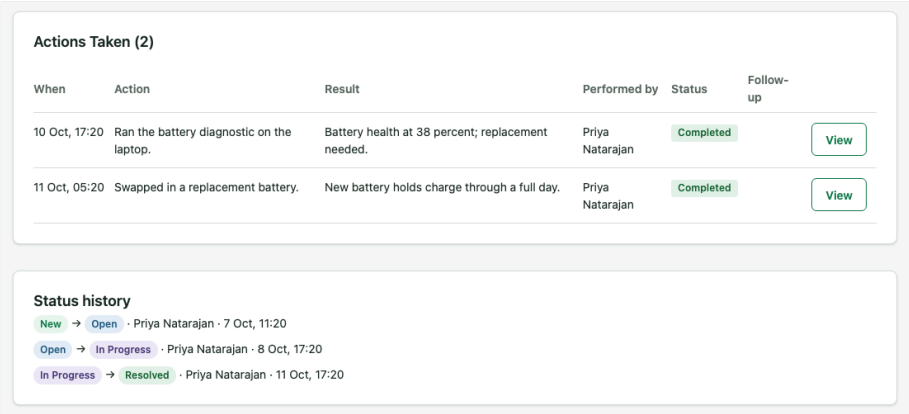
Actions Taken list on Staff Ticket Detail — desktop



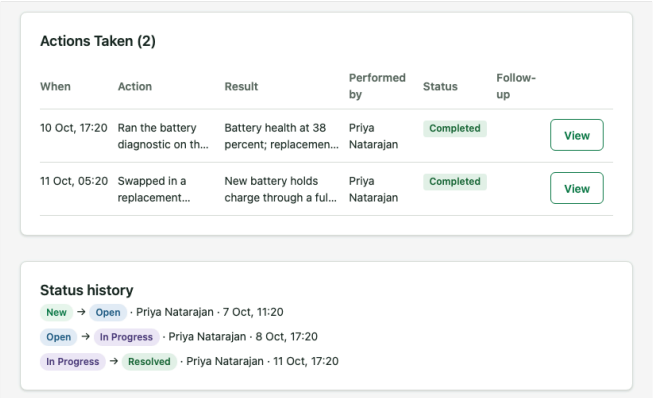
tablet



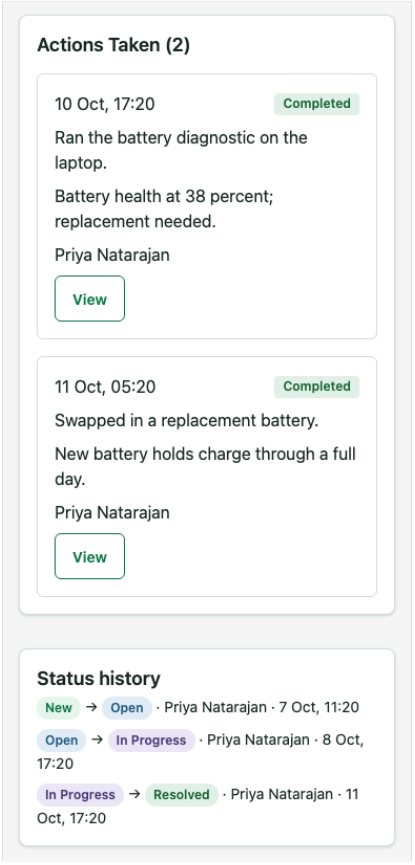
mobile



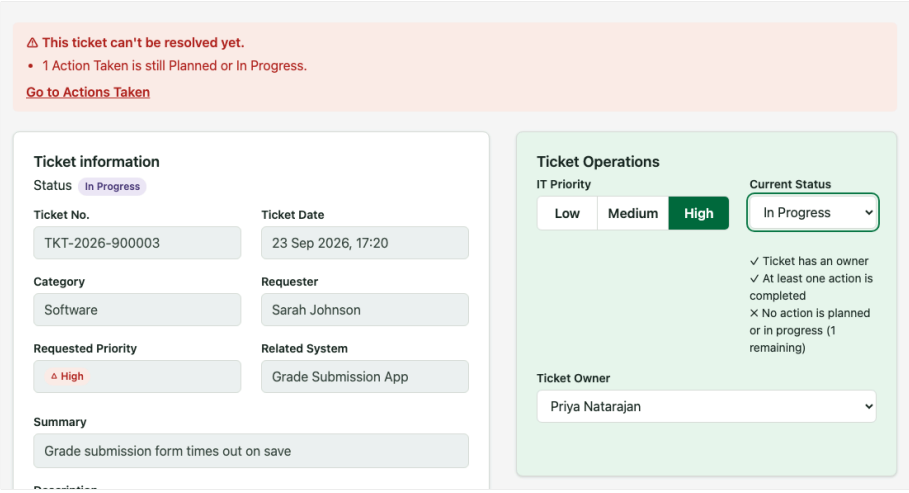
Requester read-only Actions Taken and history — desktop



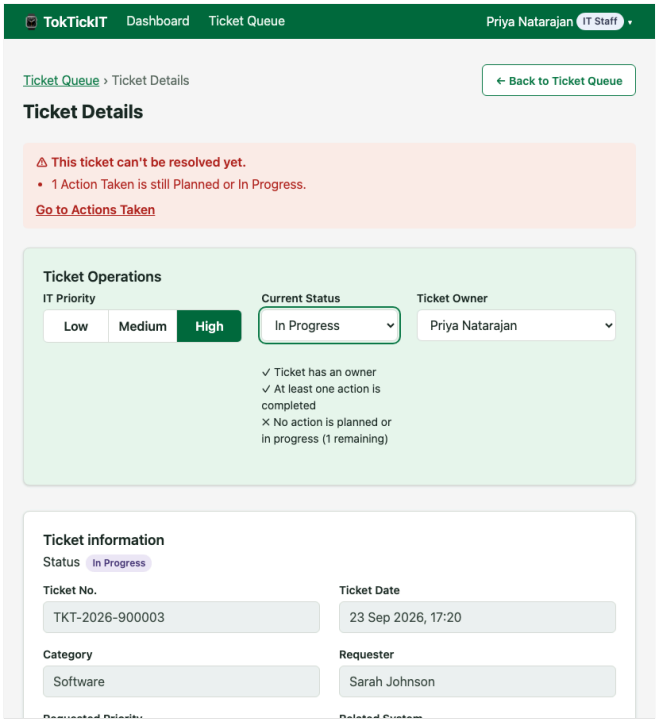
tablet



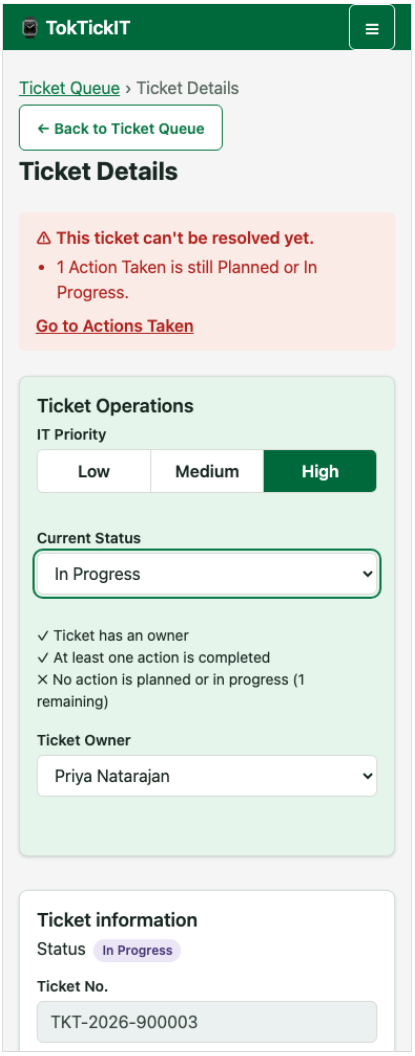
mobile



Ticket workflow: resolution blocked — desktop



tablet



mobile

Completed visual and accessibility checklist (`ui-spec.md` §10; each row is proven by the tests and screenshots in its Evidence column, and every test ID below is Pass in `tests.md` §2 on the final run):

#	Check	Evidence	Result
V-01	Every colour comes from a token; no hard-coded hex outside `theme.css`.	S-01	Pass
V-02	Dashboard, Actions Taken and history are visually of a piece with Labs 2–3 (card, spacing, type scale).	E2E-19	Pass

V-03	Dashboard is first in every role's nav, active state shown, and no unauthorized destination rendered.	C-01, E2E-15	Pass
V-04	Metric cards: label, value, sub-label and link affordance all visible; zero cards are still readable; needs-attention is conveyed by text, not colour alone.	C-03, C-08, S-03	Pass
V-05	Dashboard lists at most 5 rows, with a "View all (N)" link, and empty lists show their message.	C-04, C-06, C-09	Pass
V-06	Editable vs read-only is unmistakable in the Action form and View mode (surface vs `--zen-readonly-bg`; Performed by is static text).	C-11, C-17, S-05	Pass
V-07	Validation messages sit directly below their field; the stale and blocked callouts keep user input visible.	C-12, C-14, C-15, C-24	Pass
V-08	`ActionStatusBadge` is distinct from `StatusBadge`; Ticket status, IT Priority and Requested Priority badges stay consistent everywhere.	S-02	Pass
V-09	The status control offers only permitted transitions; the resolution checklist matches the data.	C-22, C-24, E2E-07	Pass
V-10	Visible focus on every interactive element, including metric cards, row buttons and the conditional follow-up note; logical tab order.	A11Y-05, A11Y-06	Pass
V-11	No clipping, no overlap, no hidden primary action; long text clamps with a `title`.	R-01, R-02, R-03, R-04	Pass
V-12	No horizontal page scroll at 320, 768, 992 and 1440 px.	R-01, R-02, R-03, R-04	Pass
V-13	Loading, empty, no-results, forbidden, not-found, stale, blocked and failure states each render distinctly.	C-06, C-09, C-14, C-15, C-24, C-28, C-30	Pass
V-14	Dialogs usable at mobile width and restore focus on close.	R-05, C-18	Pass
V-15	Requester Ticket Detail shows Actions Taken and history read-only with no write controls and no Internal Notes.	C-27, R-04, A11Y-04	Pass
V-16	No console errors, broken links, placeholder or lorem text, or unfinished controls on any main journey.	E2E-17	Pass
V-17	Private vs shared content is distinguishable at a glance and without colour: Internal Notes keep the private surface, border and "Private" badge; Actions Taken, status history and Public Comments carry no private styling, and the Actions Taken header states "Visible to the requester".	S-06, C-27	Pass

Note (V-12): the project's three-tier viewport matrix (390 / 820 / 1440) is used for screenshots; the responsive tests additionally assert 320, 768 and 992 px (R-01..R-04) and the 44 × 44 px drill-down chip target at 390 px (R-06).